
Romberg-Extrapolated Zeroth-Order Gradient Estimator: Higher-Order Bias Reduction with Preserved Leading Directional Variance

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Abstract

The standard two-point gradient estimator commonly used for zeroth-order optimization incurs a truncation bias of order $\mathcal{O}(r^2)$, where r is the smoothing radius. Existing bias-reduction schemes typically increase the leading directional variance under a fixed number of function evaluations, while variance-reduction schemes generally do not improve the bias order. We propose Romberg-ZOGE, a Romberg-extrapolated zeroth-order gradient estimator, which forms a weighted linear combination of two-point differences evaluated at radii $\{r/2^k\}_{k=0}^R$ while reusing the same perturbation direction across all radii. With appropriately chosen weights, Romberg-ZOGE cancels the first R even-order truncation terms and achieves $\mathcal{O}(r^{2R+2})$ bias under $(2R+2)$ -order smoothness, while preserving the leading directional variance constant of the two-point estimator up to higher-order residual terms. We further characterize the stochastic-oracle setting by deriving an explicit noise-amplification factor and establishing corresponding bias and variance bounds. Under the same number of function evaluations, Romberg-ZOGE yields faster and more stable optimization on synthetic benchmarks, simulator-based wireless network optimization, and black-box LLM prompt tuning.

1. Introduction

We study the unconstrained minimization problem

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}), \quad (1)$$

in regimes where gradients of f are unavailable or prohibitively expensive to compute. Such settings motivate zeroth-order optimization (ZOO), which relies only on function evaluations. A standard approach is to form a zeroth-order gradient estimator (ZOGE) via finite differences along random directions, and then use this estimator within a first-order method. This ZOO paradigm underlies a broad range of applications, including automated hyperparameter tuning (Chen et al., 2024), safety-critical control (Zalkind et al., 2020), and memory-constrained fine-tuning of large language models (LLMs) (Malladi et al., 2023).

The performance of this paradigm hinges on the choice of ZOGE: its bias and variance directly determine the efficiency and attainable accuracy of ZOO methods (Spall, 1992; Liu et al., 2020). We consider the standard two-point central-difference estimator as the baseline. Let $\mathbf{z}$ be an isotropic, symmetric random direction (e.g., uniform on the sphere or Gaussian); querying f at $\mathbf{x} \pm r\mathbf{z}$ with smoothing radius $r > 0$ yields the two-point central-difference estimator

$$\mathbf{g}_{2p}(\mathbf{x}; r, \mathbf{z}) \triangleq \frac{n}{2r} \left(f(\mathbf{x} + r\mathbf{z}) - f(\mathbf{x} - r\mathbf{z}) \right) \mathbf{z}. \quad (2)$$

In the deterministic setting, $\mathbf{g}_{2p}$ exhibits a canonical trade-off between truncation bias and directional variance as a function of r : the truncation bias scales as $\mathcal{O}(r^2)$, while the leading directional variance term scales as $V \|\nabla f(\mathbf{x})\|^2$ (up to $\mathcal{O}(r^2)$ remainders), where V depends on the direction distribution and the dimension. Consequently, decreasing r reduces the bias but does not reduce the dominant variance term. Moreover, in many black-box applications r cannot be made arbitrarily small due to finite numerical resolution, so the $\mathcal{O}(r^2)$ truncation bias becomes a practical accuracy barrier when r is bounded away from zero.

Limitations of existing approaches. To improve ZOO, prior work typically targets either the variance or the bias of the two-point estimator. Variance-reduction methods (e.g., mini-batching (Duchi et al., 2015; Liu et al., 2018a)

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or structured directions (Kozak et al., 2023; Ma & Huang, 2025a)) aim to decrease the variance constant V but generally leave the $\mathcal{O}(r^2)$ truncation bias unchanged. In contrast, bias-reduction methods (e.g., kernel-based estimators (Bach & Perchet, 2016) or randomized telescoping (Chen, 2020; Ma & Huang, 2025b)) reduce the truncation bias to higher order, but often increase the leading directional variance constant when the number of function evaluations per gradient estimate is fixed.

A central question. These limitations motivate the following question in the deterministic setting.

Q1: Can we reduce the truncation bias without increasing the leading directional variance constant V ?

We answer **Q1** affirmatively by proposing **Romberg-ZOGE**, a Romberg-extrapolated zeroth-order gradient estimator. The key idea is to evaluate the two-point estimator at a geometric sequence of radii $r_k \triangleq r/2^k$ and form a weighted linear combination, while reusing the same direction $\mathbf{z}$ across all radii. With appropriately chosen weights, the first R even-order truncation terms cancel, improving the bias from $\mathcal{O}(r^2)$ to $\mathcal{O}(r^{2R+2})$. Crucially, sharing $\mathbf{z}$ across radii enables pathwise cancellation of the dominant directional randomness, so the leading directional variance constant matches that of g_{2p} (up to higher-order residual terms).

Contributions. Our main contributions are:

1. **Estimator construction.** We propose **Romberg-ZOGE**, a Romberg-extrapolated combination of two-point estimators evaluated at radii $\{r/2^k\}_{k=0}^R$ using a shared direction $\mathbf{z}$.
2. **Bias and variance theory under deterministic and stochastic oracles.** Under $(2R+2)$ -order smoothness, Romberg-ZOGE attains truncation bias $\mathcal{O}(r^{2R+2})$ while preserving the leading directional variance constant V of the two-point estimator, up to higher-order residual terms. We further extend the analysis to a stochastic oracle. Specifically, we derive an explicit factor that quantifies noise amplification and establish corresponding bounds on the bias and variance.
3. **Empirical validation.** Under the same number of function evaluations per gradient estimate, Romberg-ZOGE yields faster and more stable optimization on synthetic benchmarks, simulator-based wireless network optimization, and black-box LLM prompt tuning.

2. Background

2.1. Preliminaries

We summarize the notation and assumptions used throughout the paper. Unless stated otherwise, we consider deterministic function evaluations; the stochastic-oracle extension is deferred to Section 4.2.

Direction distributions. We use direction distributions that are isotropic and symmetric, which ensures $\mathbf{z} \stackrel{d}{=} -\mathbf{z}$ and $\mathbb{E}[\mathbf{z}\mathbf{z}^\top] = \mathbf{I}_n/n$.

Definition 2.1 (Direction distributions). *We consider two isotropic and symmetric distributions for $\mathbf{z}$:*

- **Uniform spherical:** $\mathbf{z} \sim \text{Unif}(\mathbb{S}^{n-1})$, i.e., $\mathbf{z}$ is sampled uniformly at random from the unit sphere.
- **Gaussian:** $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$, i.e., $\mathbf{z}$ is a zero-mean Gaussian vector with covariance $\mathbf{I}_n/n$.

Assumption 2.1 (High-order smoothness). *For an integer $\beta \geq 1$, the objective $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is β -times continuously differentiable. Moreover, the gradient ∇f and the β -th derivative tensor $\nabla^\beta f$ are Lipschitz in operator norm:*

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L_1 \|\mathbf{x} - \mathbf{y}\|, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \quad (3)$$

$$\|\nabla^\beta f(\mathbf{x}) - \nabla^\beta f(\mathbf{y})\|_{\text{op}} \leq L_\beta \|\mathbf{x} - \mathbf{y}\|, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n. \quad (4)$$

Remark 2.1 (Scope of Assumption 2.1). *Assumption 2.1 is standard for analyzing high-order finite-difference and extrapolation schemes. While modern neural networks may not satisfy such conditions globally, they are often locally smooth in a neighborhood of the query point; our results can be interpreted under such local regularity.*

Bias and variance of the two-point estimator. Under Assumption 2.1 with $\beta = 2$, we measure accuracy and variability only with respect to the random direction $\mathbf{z}$. For any estimator $g(\mathbf{x}; r, \mathbf{z}) \in \mathbb{R}^n$, define $\text{Var}_{\mathbf{z}}(g(\mathbf{x}; r, \mathbf{z})) \triangleq \mathbb{E}_{\mathbf{z}} \|g(\mathbf{x}; r, \mathbf{z}) - \mathbb{E}_{\mathbf{z}} g(\mathbf{x}; r, \mathbf{z})\|^2$. The two-point estimator in (2) then satisfies the following standard bounds (Nesterov & Spokoiny, 2017; Duchi et al., 2015).

Lemma 2.1 (Bias and variance of the two-point ZOGE). *For $\mathbf{z}$ drawn from either distribution in Definition 2.1,*

$$\|\mathbb{E}_{\mathbf{z}}[g_{2p}(\mathbf{x}; r, \mathbf{z})] - \nabla f(\mathbf{x})\| = \mathcal{O}(r^2), \quad (5)$$

$$\text{Var}_{\mathbf{z}}(g_{2p}(\mathbf{x}; r, \mathbf{z})) = V \|\nabla f(\mathbf{x})\|^2 + \mathcal{O}(r^2), \quad (6)$$

where $V = n - 1$ if $\mathbf{z} \sim \text{Unif}(\mathbb{S}^{n-1})$ and $V = n + 1$ if $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$.

2.2. Related Work and Gap

We review prior approaches to bias and variance reduction for ZOGEs.

Variance reduction. A first line of work aims to reduce the directional variance of ZOGES. Common strategies include averaging over multiple random directions (mini-batching (Duchi et al., 2015; Liu et al., 2018a)) and adapting variance-reduction techniques from first-order optimization to the zeroth-order setting (e.g., ZO-SVRG (Liu et al., 2018b)). More recently, structured directions have been explored to reduce the constant V in (6) while retaining the two-point form (Kozak et al., 2023; Ma & Huang, 2025a). These methods primarily reduce the coefficient multiplying $\|\nabla f(\mathbf{x})\|^2$, but typically do not improve the $\mathcal{O}(r^2)$ truncation bias inherent to two-point finite differences.

Bias reduction. A second line of work aims to reduce truncation bias beyond $\mathcal{O}(r^2)$. Kernel-based methods construct higher-order smoothing kernels that cancel low-order Taylor terms in expectation (Bach & Perchet, 2016). Multi-point estimators (Li et al., 2022) query the function at multiple points along each coordinate direction to obtain higher-order bias, but their per-iteration evaluation cost scales linearly with the dimension, making them prohibitive in high-dimensional settings. Randomized telescoping techniques introduce a random truncation level to obtain unbiased estimators in expectation (Chen, 2020; Ma & Huang, 2025b). However, these bias improvements often come with increased variance under a fixed number of function evaluations, which can limit their effectiveness in optimization.

Gap. To the best of our knowledge, in the deterministic setting no existing method reduces the truncation bias from $\mathcal{O}(r^2)$ to $\mathcal{O}(r^{2R+2})$ (for $R \geq 1$) while preserving the leading directional variance constant V of the standard two-point ZOGES under a fixed number of function evaluations per gradient estimate. Romberg-ZOGES fills this gap by exploiting the even-power structure of central differences and by reusing the same perturbation direction across radii to enable pathwise cancellation. As a result, the first R even-order truncation terms are eliminated by design, while the leading directional variance constant is preserved.

3. Romberg-Extrapolated ZOGES

This section presents Romberg-ZOGES, a Romberg-extrapolated zeroth-order gradient estimator. Starting from the two-point estimator (2), we combine finite-difference estimates computed at multiple smoothing radii to cancel low-order truncation terms, while reusing a single perturbation direction across radii to preserve the leading directional variance constant. As shown in Section 4, this design improves the truncation bias to $\mathcal{O}(r^{2R+2})$ without inflating the leading directional variance term of the two-point estimator.

Algorithm 1 Romberg-ZOGES

- 1: **Input:** current point $\mathbf{x}$, base radius r , order R , direction distribution $\mathcal{D}(\mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$ or $\text{Unif}(\mathbb{S}^{n-1}))$.
 - 2: **Precompute:** weights $\{w_k^{(R)}\}_{k=0}^R$ via (12).
 - 3: Sample one direction $\mathbf{z} \sim \mathcal{D}$.
 - 4: Initialize $\mathbf{g}^{[R]} \leftarrow \mathbf{0}$.
 - 5: **for** $k = 0$ **to** R **do**
 - 6: Set $r_k \leftarrow r/2^k$.
 - 7: Query $f(\mathbf{x} + r_k \mathbf{z})$ and $f(\mathbf{x} - r_k \mathbf{z})$.
 - 8: Compute $\mathbf{g}_k \leftarrow \frac{n}{2r_k} (f(\mathbf{x} + r_k \mathbf{z}) - f(\mathbf{x} - r_k \mathbf{z})) \mathbf{z}$.
 - 9: Update $\mathbf{g}^{[R]} \leftarrow \mathbf{g}^{[R]} + w_k^{(R)} \mathbf{g}_k$.
 - 10: **end for**
 - 11: **Return:** $\mathbf{g}^{[R]}$.
-

3.1. Estimator Definition

We evaluate the two-point estimator (2) at a geometric sequence of radii

$$r_k \triangleq \frac{r}{2^k}, \quad k = 0, 1, \dots, R, \quad (7)$$

and define the Romberg-extrapolated estimator

$$\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z}) \triangleq \sum_{k=0}^R w_k^{(R)} \mathbf{g}_{2^k}(\mathbf{x}; r_k, \mathbf{z}), \quad (8)$$

where the weights $\{w_k^{(R)}\}_{k=0}^R$ are chosen to cancel the first R even-order truncation terms; see Section 3.2. A key design choice is to reuse the same sampled direction $\mathbf{z}$ for all radii $\{r_k\}_{k=0}^R$. This shared-direction construction enables pathwise cancellation of the dominant directional randomness, which is crucial for preserving the leading directional variance constant. Algorithm 1 summarizes the procedure.

3.2. Weight Design via Bias Cancellation

We now choose $\{w_k^{(R)}\}_{k=0}^R$ to cancel low-order truncation terms in $\mathbb{E}_{\mathbf{z}}[\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z})]$. For symmetric perturbations and central differences, the bias admits an expansion in even powers of the smoothing radius, which is precisely the structure exploited by Romberg extrapolation.

Assume Assumption 2.1 holds with $\beta = 2R + 2$, and let $\mathbf{z} \sim \mathcal{D}$ where $\mathcal{D}$ is either $\text{Unif}(\mathbb{S}^{n-1})$ or $\mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$. Then the two-point estimator satisfies the expansion

$$\mathbb{E}_{\mathbf{z}}[\mathbf{g}_{2^k}(\mathbf{x}; r, \mathbf{z})] = \nabla f(\mathbf{x}) + \sum_{j=1}^R c_{2j}(\mathbf{x}) r^{2j} + \mathcal{O}(r^{2R+2}), \quad (9)$$

where $\{c_{2j}(\mathbf{x})\}_{j \geq 1}$ are deterministic vectors depending on f and $\mathcal{D}$ but not on r ; see Section A.

Substituting (7) into (8) and applying (9) yields

$$\begin{aligned} \mathbb{E}_z \left[\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z}) \right] &= \left(\sum_{k=0}^R w_k^{(R)} \right) \nabla f(\mathbf{x}) + \mathcal{O}(r^{2R+2}) \\ &\quad + \sum_{j=1}^R c_{2j}(\mathbf{x}) r^{2j} \left(\sum_{k=0}^R w_k^{(R)} 4^{-jk} \right). \end{aligned} \quad (10)$$

To preserve the coefficient of $\nabla f(\mathbf{x})$ and eliminate the coefficients of r^{2j} for $j = 1, \dots, R$, we choose $\{w_k^{(R)}\}$ to satisfy

$$\sum_{k=0}^R w_k^{(R)} = 1, \quad \sum_{k=0}^R w_k^{(R)} 4^{-jk} = 0, \quad j = 1, \dots, R. \quad (11)$$

This is a square Vandermonde-type system and has a unique solution.

Lemma 3.1 (Romberg weights). *The system (11) has a unique solution given by*

$$w_k^{(R)} = \prod_{\substack{j=0 \\ j \neq k}}^R \frac{1}{1 - 4^{j-k}}, \quad k = 0, 1, \dots, R. \quad (12)$$

Remark 3.1 (Function-evaluation cost). *Computing $\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z})$ requires $2(R+1)$ function evaluations.*

4. Theory

This section provides the theoretical analysis of Romberg-ZOGE. We first establish two estimator-level guarantees under a deterministic oracle: (i) Romberg-ZOGE improves the truncation bias to order $\mathcal{O}(r^{2R+2})$, and (ii) by reusing the same direction across radii, the estimator preserves the leading directional variance constant of the two-point estimator through pathwise cancellation. We then extend the analysis to a stochastic oracle and quantify the resulting noise amplification. Finally, we translate these estimator properties into convergence and complexity guarantees for zeroth-order stochastic gradient descent (ZO-SGD).

4.1. Deterministic Oracle: Bias and Directional Variance

We begin with the deterministic setting, where the oracle returns $f(\mathbf{x})$ exactly for any queried point $\mathbf{x}$ and the only randomness comes from the sampled direction $\mathbf{z}$.

First, we quantify the resulting high-order bias reduction in the following theorem.

Theorem 4.1 (Bias of Romberg-ZOGE). *Assume Assumption 2.1 holds with $\beta = 2R+2$. Let $\mathbf{z} \sim \mathcal{D}$, where $\mathcal{D}$ is either $\text{Unif}(\mathbb{S}^{n-1})$ or $\mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$. Define $\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z})$ by (8) with weights satisfying (11). Then, for all $\mathbf{x} \in \mathbb{R}^n$ and*

all $r \in (0, 1]$,

$$\left\| \mathbb{E}_z \left[\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z}) \right] - \nabla f(\mathbf{x}) \right\| \leq c_b^{(R)} r^{2R+2}, \quad (13)$$

where $c_b^{(R)}$ depends on $n, R, \mathcal{D}$, and L_{2R+2} in Assumption 2.1, but is independent of $\mathbf{x}$ and r .

We next show that this bias reduction does not inflate the dominant directional variance term: Romberg-ZOGE matches the leading variance constant of the two-point estimator, up to higher-order residual terms.

Theorem 4.2 (Variance of Romberg-ZOGE). *Assume Assumption 2.1 holds with $\beta = 2R+2$. Let $\mathbf{z} \sim \mathcal{D}$, where $\mathcal{D}$ is either $\text{Unif}(\mathbb{S}^{n-1})$ or $\mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$. Then for all $\mathbf{x} \in \mathbb{R}^n$ and all $r \in (0, 1]$,*

$$\begin{aligned} \text{Var}_z \left(\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z}) \right) &\leq V \|\nabla f(\mathbf{x})\|^2 + c_{v,1}^{(R)} \|\nabla f(\mathbf{x})\| r^{2R+2} + c_{v,2}^{(R)} r^{4R+4}, \end{aligned} \quad (14)$$

where $V = n-1$ for $\mathcal{D} = \text{Unif}(\mathbb{S}^{n-1})$ and $V = n+1$ for $\mathcal{D} = \mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$. The constants $c_{v,1}^{(R)}$ and $c_{v,2}^{(R)}$ depend on $n, R, \mathcal{D}$, and L_{2R+2} in Assumption 2.1, but are independent of $\mathbf{x}$ and r .

Theorem 4.2 shows that Romberg-ZOGE preserves the dominant directional-variance term $V \|\nabla f(\mathbf{x})\|^2$ of the two-point estimator, while the remaining variance terms vanish as $r \rightarrow 0$.

Remark 4.1 (Why reusing the same direction preserves the leading variance). *If the same direction $\mathbf{z}$ is reused across radii, then the leading term is shared and the dominant variance scales as $V \left(\sum_{k=0}^R w_k^{(R)} \right)^2 = V$ since $\sum_{k=0}^R w_k^{(R)} = 1$. If independent directions $\{\mathbf{z}_k\}_{k=0}^R$ are used instead, the leading terms add in variance, yielding $V \sum_{k=0}^R (w_k^{(R)})^2 \|\nabla f(\mathbf{x})\|^2$ and thus an inflation factor $\sum_{k=0}^R (w_k^{(R)})^2 \geq 1$.*

4.2. Stochastic Oracle: Noisy Function Evaluations

We now consider stochastic settings where exact function values are unavailable and we can only query a noisy oracle $\tilde{f}(\mathbf{x})$. In this regime, Romberg-ZOGE preserves the higher-order truncation-bias reduction, but it can amplify oracle noise. We formalize the oracle model below.

Assumption 4.1 (Stochastic oracle). *For any query point $\mathbf{x} \in \mathbb{R}^n$, the oracle returns $\tilde{f}(\mathbf{x}; \xi)$ such that:*

- **Unbiasedness:** $\mathbb{E}_\xi[\tilde{f}(\mathbf{x}; \xi)] = f(\mathbf{x})$ for all $\mathbf{x}$.
- **Bounded variance:** $\text{Var}_\xi(\tilde{f}(\mathbf{x}; \xi)) \leq \sigma^2 < \infty$ for all $\mathbf{x}$.
- **Independence:** Oracle noises are independent across different query points: for any distinct $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $\tilde{f}(\mathbf{x}; \xi_x)$ and $\tilde{f}(\mathbf{y}; \xi_y)$ are independent, where ξ_x and ξ_y are i.i.d.

noise samples. Moreover, for all $\mathbf{x}$, $\tilde{f}(\mathbf{x}; \xi)$ is independent of the direction $\mathbf{z}$.

Here, ξ denotes the randomness of a single oracle call. Different oracle calls use independent copies of ξ , as in the independent noisy function-value oracle model considered in (Akhavan et al., 2024).

Remark 4.2 (Examples of stochastic oracle). *Assumption 4.1 includes (i) stochastic objectives $f(\mathbf{x}) = \mathbb{E}_\xi[\tilde{f}(\mathbf{x}; \xi)]$, where each query returns one sample $\tilde{f}(\mathbf{x}; \xi)$, and (ii) additive noise $\tilde{f}(\mathbf{x}; \xi) = f(\mathbf{x}) + \mathbf{n}(\xi)$ with $\mathbb{E}_\xi[\mathbf{n}(\xi)] = 0$ and $\text{Var}_\xi(\mathbf{n}(\xi)) \leq \sigma^2$.*

Let $\tilde{\mathbf{g}}^{[R]}(\mathbf{x}; r, \mathbf{z})$ denote Romberg-ZOGE obtained by replacing f with $\tilde{f}$ in Algorithm 1. Denote the $2(R+1)$ oracle noises in one estimator as $\xi^{[R]} = \{\xi_{k,+}, \xi_{k,-}\}_{k=0}^R$. The next result shows that the bias bound is unchanged, while the variance acquires an additional term capturing noise accumulation.

Theorem 4.3 (Bias and variance with a stochastic oracle). *Suppose Assumption 2.1 holds with $\beta = 2R+2$ and Assumption 4.1 holds. Let $\mathbf{z} \sim \mathcal{D}$, where $\mathcal{D}$ is either $\text{Unif}(\mathbb{S}^{n-1})$ or $\mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$. Then, for all $\mathbf{x} \in \mathbb{R}^n$ and all $r \in (0, 1]$, the bias satisfies*

$$\left\| \mathbb{E}_{\mathbf{z}, \xi^{[R]}} [\tilde{\mathbf{g}}^{[R]}(\mathbf{x}; r, \mathbf{z})] - \nabla f(\mathbf{x}) \right\| \leq c_b^{(R)} r^{2R+2}, \quad (15)$$

and the variance satisfies

$$\begin{aligned} & \text{Var}_{\mathbf{z}, \xi^{[R]}} (\tilde{\mathbf{g}}^{[R]}(\mathbf{x}; r, \mathbf{z})) \\ & \leq V \|\nabla f(\mathbf{x})\|^2 + c_{v,1}^{(R)} \|\nabla f(\mathbf{x})\| r^{2R+2} \\ & + c_{v,2}^{(R)} r^{4R+4} + \frac{n^2 \sigma^2}{2r^2} \kappa_R, \end{aligned} \quad (16)$$

where V , $c_b^{(R)}$, $c_{v,1}^{(R)}$, and $c_{v,2}^{(R)}$ are defined as in Theorem 4.2, and $\kappa_R \triangleq \sum_{k=0}^R (w_k^{(R)})^2 4^k$.

Remark 4.3 (Noise amplification and choice of R). *Under a stochastic oracle, larger R reduces truncation bias but increases the noise term through κ_R . In practice, R should be chosen to balance these effects.*

4.3. Convergence and Complexity with ZO-SGD

Having established estimator-level bias and variance guarantees, we now translate them into optimization convergence rates for $\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$ under ZO-SGD. Let $\hat{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t)$ denote the estimator used at iteration t : it equals $\mathbf{g}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t)$ under a deterministic oracle and $\tilde{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t)$ under a stochastic oracle.

The update is

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \hat{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t), \quad (17)$$

where η is the stepsize and $\{\mathbf{z}_t\}_{t \geq 0}$ are i.i.d. directions drawn from $\mathcal{D} \in \{\text{Unif}(\mathbb{S}^{n-1}), \mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)\}$.

For each iteration t , let $\xi_t^{[R]}$ collect all oracle noises used to form $\hat{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t)$. Define the history sigma-algebra

$$\Omega_t \triangleq \sigma(\mathbf{z}_0, \dots, \mathbf{z}_t, \xi_0^{[R]}, \dots, \xi_t^{[R]}), \quad t \geq 0, \quad (18)$$

and write $\mathbb{E}_{\Omega_t}[\cdot]$ for expectation with respect to all randomness in Ω_t .

To state a standard non-convex guarantee, we impose a mild lower-boundedness condition.

Assumption 4.2 (Lower boundedness). *There exists $f^* > -\infty$ such that $f(\mathbf{x}) \geq f^*$ for all $\mathbf{x} \in \mathbb{R}^n$.*

Under Assumption 4.2, we obtain a standard nonconvex stationarity guarantee.

Theorem 4.4 (Nonconvex convergence). *Suppose Assumptions 4.2 and 2.1 hold with $\beta = 2R+2$, and Assumption 4.1 holds. Run ZO-SGD (17) with estimator $\hat{\mathbf{g}}^{[R]}$ and stepsize $0 < \eta \leq \frac{1}{8L_1(V+1)}$. Then for any $T \geq 1$ and any $r \in (0, 1]$, we have*

$$\begin{aligned} & \min_{0 \leq t \leq T-1} \mathbb{E}_{\Omega_{T-1}} \|\nabla f(\mathbf{x}_t)\|^2 \\ & \leq \frac{2(f(\mathbf{x}_0) - f^*)}{\eta T} + L_1 \eta \cdot \frac{n^2 \sigma^2}{2r^2} \kappa_R + C_R r^{4R+4}, \end{aligned} \quad (19)$$

where κ_R is defined in Theorem 4.3, and $C_R > 0$ depends on $n, R, \mathcal{D}, L_1, V$ and the estimator constants in Theorem 4.3, but is independent of T, η, r .

As a direct consequence of (19), we obtain the following iteration-complexity corollary.

Corollary 4.1 (Iteration Complexity). *Under the assumptions of Theorem 4.4, the iteration complexity to reach an ε -stationary point (i.e., $\min_{0 \leq t \leq T-1} \mathbb{E}_{\Omega_{T-1}} \|\nabla f(\mathbf{x}_t)\|^2 \leq \varepsilon^2$) satisfies:*

1. **Deterministic oracle** ($\sigma = 0$). *Choose any constant stepsize $0 < \eta \leq \frac{1}{8L_1(V+1)}$ and smoothing radius $r \leq \left(\frac{\varepsilon^2}{3C_R}\right)^{\frac{1}{4R+4}}$, then the iteration complexity is $T = \mathcal{O}(\varepsilon^{-2})$.*
2. **Stochastic oracle** ($\sigma > 0$). *Choose $r \leq \left(\frac{\varepsilon^2}{3C_R}\right)^{\frac{1}{4R+4}}$ and $\eta \leq \min \left\{ \frac{1}{8L_1(V+1)}, \frac{2\varepsilon^2 r^2}{3L_1 n^2 \sigma^2 \kappa_R} \right\}$, then the iteration complexity is $T = \mathcal{O}(\varepsilon^{-(4 + \frac{1}{R+1})})$.*

Remark 4.4. *In the deterministic oracle case, the iteration complexity $T = \mathcal{O}(\varepsilon^{-2})$ matches the standard deterministic-oracle rate (Nesterov & Spokoiny, 2017).*

Remark 4.5 (Stochastic-oracle models and comparison with prior rates). *The stochastic-oracle result in Theorem 4.3 and Corollary 4.1 assumes independent noisy function-value evaluations at the $2(R + 1)$ queried points. This model introduces the noise-amplification factor κ_R/r^2 , yielding the complexity $\mathcal{O}(\varepsilon^{-(4+1/(R+1))})$. Hence, the rate is a consequence of the independent-noise oracle model, not a limitation of the Romberg construction itself.*

In contrast, under the shared-sample stochastic-objective model commonly used in prior zeroth-order work, the same sample ξ is reused across all queried points within one estimator; for Romberg-ZOGE this means evaluating $\tilde{f}(\mathbf{x} \pm r_k \mathbf{z}; \xi)$, $k = 0, \dots, R$, with a common ξ . Under the standard bounded stochastic-gradient variance condition

$$\mathbb{E}_\xi \|\nabla \tilde{f}(\mathbf{x}; \xi) - \nabla f(\mathbf{x})\|^2 \leq \sigma_g^2,$$

the κ_R/r^2 amplification vanishes. As a result, the corresponding ZO-SGD guarantee becomes

$$\begin{aligned} & \min_{0 \leq t \leq T-1} \mathbb{E} \|\nabla f(\mathbf{x}_t)\|^2 \\ & \leq \frac{2(f(\mathbf{x}_0) - f^*)}{\eta T} + \tilde{C}_1 n \eta \sigma_g^2 + \tilde{C}_R r^{4R+4}, \end{aligned} \quad (20)$$

with constants $\tilde{C}_1, \tilde{C}_R$ independent of T, η, r, σ_g , and ε . Choosing $r \asymp \varepsilon^{1/(2R+2)}$ and $\eta \asymp \varepsilon^2/(n\sigma_g^2)$ gives $T = \mathcal{O}(n\sigma_g^2\varepsilon^{-4})$, matching (Duchi et al., 2015) up to the same σ_g^2 factor. The detailed formulation and proof are deferred to Appendix E.1. Additional noisy R -sensitivity experiments under both independent-noise and shared-sample stochastic formulations are reported in Appendix F.3.

5. Experiments

We evaluate Romberg-ZOGE from three perspectives. First, we validate estimator-level properties, focusing on high-order truncation-bias decay and preservation of the leading directional variance constant. Second, we study optimization performance by integrating each estimator into ZO-SGD and comparing convergence when the number of function evaluations per iteration is fixed. Third, we assess practical performance on application benchmarks, including simulator-based wireless network optimization and black-box LLM prompt tuning.

5.1. Baselines

Throughout this section, we use Romberg-ZOGE with $R = 2$, which achieves $\mathcal{O}(r^6)$ truncation bias. We compare against representative ZOGEs: the standard two-point estimator, a kernel-based estimator (Bach & Perchet, 2016), and the unbiased estimator P4 (Ma & Huang, 2025b). Full definitions of all baselines are provided in Section F.1.

Each gradient estimate uses $Q = 2$ function evaluations

for Two-point and Kernel, $Q = 4$ for P4, and $Q = 6$ for Romberg-ZOGE with $R = 2$.

5.2. Synthetic Validation

We first validate the theory on smooth synthetic objectives and then examine whether the estimator-level gains translate into faster optimization when the number of function evaluations per iteration is fixed. Throughout this subsection, we set $n = 100$ and report results over 10 random initializations.

Test functions. We consider two objectives capturing polynomial growth and nonconvex geometry:

- **Poly8:** $f_{\text{poly}}(\mathbf{x}) = \frac{1}{8} \sum_{i=1}^n x_i^8$.
- **Rosenbrock:** $f_{\text{rb}}(\mathbf{x}) = \sum_{i=1}^{n-1} \left(100(x_{i+1} - x_i^2)^2 + (1 - x_i)^2 \right)$.

Bias validation. We verify the truncation-bias order predicted by Theorem 4.1. We estimate the bias term as

$$\text{Bias}(r) \triangleq \frac{1}{M} \sum_{i=1}^M \left\| \frac{1}{M_z} \sum_{j=1}^{M_z} g(\mathbf{x}_i; r, \mathbf{z}_j) - \nabla f(\mathbf{x}_i) \right\|,$$

where $\{\mathbf{z}_j\}_{j=1}^{M_z}$ are i.i.d. directions drawn from the specified distribution. For each objective, we sample $M = 10$ random points $\{\mathbf{x}_i\}_{i=1}^M$ and average across them. For each $(\mathbf{x}_i, r)$, we set $M_z = 10^7$ to approximate the directional expectation, and we reuse the same set of directions across methods when computing the directional mean.

Figure 1 compares Two-point and Romberg-ZOGE ($R = 2$) under $\text{Unif}(\mathbb{S}^{n-1})$, together with Kernel and P4. Romberg-ZOGE follows the predicted $\mathcal{O}(r^6)$ trend, whereas Two-point exhibits $\mathcal{O}(r^2)$ scaling. P4 decays more slowly and Kernel is nearly flat, suggesting a bias plateau. Since the bias is defined through an expectation over directions, it is unaffected by whether Romberg-ZOGE reuses a shared direction or samples independent directions across radii. This design choice, however, is essential for variance, which we validate next.

Remark 5.1 (Interpretation of the fitted slopes in Figure 1). *The fitted slopes in Figure 1 are empirical log-log slopes over a finite range of smoothing radii, and are intended as qualitative indicators of bias decay rather than exact asymptotic exponents. Since the bias is estimated by Monte Carlo averaging, residual sampling error, floating-point precision, and cancellation effects can affect the estimates, especially for small r . These effects explain the mild deviation of the two-point estimator from its nominal $\mathcal{O}(r^2)$ trend. For Kernel and P4, the deviation can be larger because their auxiliary randomness introduces additional Monte Carlo*

variance. A refined auxiliary-averaging experiment in Appendix F.2.2 further supports this interpretation.

Variance validation. We quantify directional variability and isolate the effect of sharing directions across radii. To validate Theorem 4.2, we measure the normalized directional variance coefficient

$$V(r) \triangleq \frac{1}{M} \sum_{i=1}^M \frac{\frac{1}{M_z} \sum_{j=1}^{M_z} \|g(\mathbf{x}_i; r, \mathbf{z}_j) - \bar{g}(\mathbf{x}_i; r)\|^2}{\|\nabla f(\mathbf{x}_i)\|^2},$$

where $\bar{g}(\mathbf{x}_i; r) \triangleq \frac{1}{M_z} \sum_{j=1}^{M_z} g(\mathbf{x}_i; r, \mathbf{z}_j)$. We use the same protocol as in the bias study (the same M and M_z , and the same set of directions across methods for each $(\mathbf{x}_i, r)$).

Figure 2 reports $V(r)$ across estimators. Romberg-ZOGE ($R = 2$) with a shared direction matches the leading directional variance coefficient of the two-point estimator. We also include an ablation, **Romberg (indep dirs)**, which samples independent directions across radii instead of reusing a shared one. As predicted by Remark 4.1, this removes pathwise cancellation and inflates the leading directional variance by $\sum_{k=0}^R (w_k^{(R)})^2$ (for $R = 2$, approximately 2.22), consistent with the figure. In contrast, Kernel and P4 yield substantially larger $V(r)$, indicating variance inflation when targeting higher-order bias reduction. Overall, these results confirm that Romberg-ZOGE preserves the dominant directional variance only when the same direction is shared across radii.

Optimization with ZO-SGD. We evaluate optimization performance by plugging each estimator into ZO-SGD while holding the number of function evaluations per iteration fixed. We set the total number of function evaluations per iteration to 120 and choose the number of sampled directions m accordingly: $m = 60$ for Two-point and Kernel ($Q = 2$), $m = 20$ for Romberg-ZOGE with $R = 2$ ($Q = 6$), and $m = 30$ for P4 ($Q = 4$). For each objective, we run ZO-SGD from the same 10 initializations and report the mean objective value versus iterations (with variability across runs). All methods use step sizes selected via the same short-horizon tail-averaged tuning protocol (Section F.2.3).

Figure 3 shows that Romberg-ZOGE converges fastest under spherical directions. Two-point converges more slowly, P4 underperforms Two-point, and Kernel makes little progress, consistent with its nearly flat bias in Figure 1. These trends align with the estimator-level results: Romberg-ZOGE and Two-point have comparable leading directional variance coefficients (Figure 2), while Romberg-ZOGE achieves substantially smaller truncation bias (Figure 1), which translates into faster optimization in this deterministic setting.

5.3. Wireless Network Optimization

We evaluate the proposed estimators on a simulator-based wireless network optimization task. In modern 5G systems, spectral efficiency (SE) depends on configurable base-station parameters such as antenna downtilt and azimuth. Following recent works (Zhang et al., 2025; Cao et al., 2022), we treat an external wireless network simulator as a strict query-only oracle that maps these parameters to location-wise SE values, with gradients unavailable. To emphasize reliability across locations, we optimize a smooth quantile objective, yielding a realistic deterministic black-box benchmark for zeroth-order methods.

Problem. Let $\mathbf{x} \in \mathbb{R}^{2Q}$ denote the downtilt and azimuth parameters for Q base stations. Given L grid locations, the evaluator returns SE values $\eta^l(\mathbf{x}) \in \mathbb{R}_0^+$ for $l = 1, \dots, L$. We maximize the smooth τ -quantile objective from (Zhang et al., 2025):

$$\max_{\mathbf{x} \in \mathbb{R}^{2Q}} f(\mathbf{x}) \triangleq \tilde{Q}_\tau \left(\{\eta^l(\mathbf{x})\}_{l=1}^L \right), \quad (21)$$

where $\tilde{Q}_\tau(\cdot)$ is the smooth quantile operator defined in Section F.4. We set $\tau = 0.9$ to emphasize reliability by focusing on the lower tail of location-wise SE.

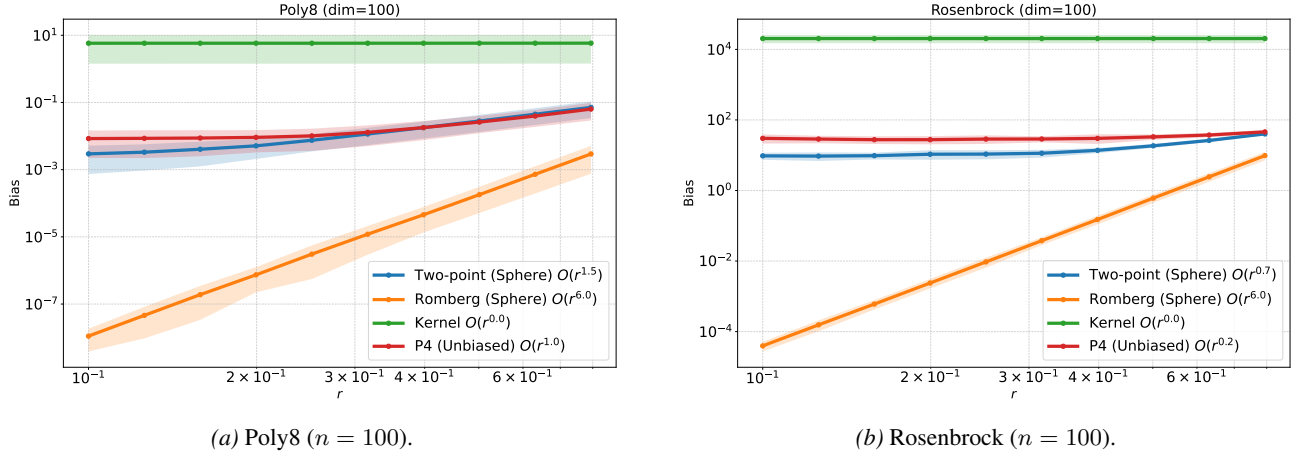
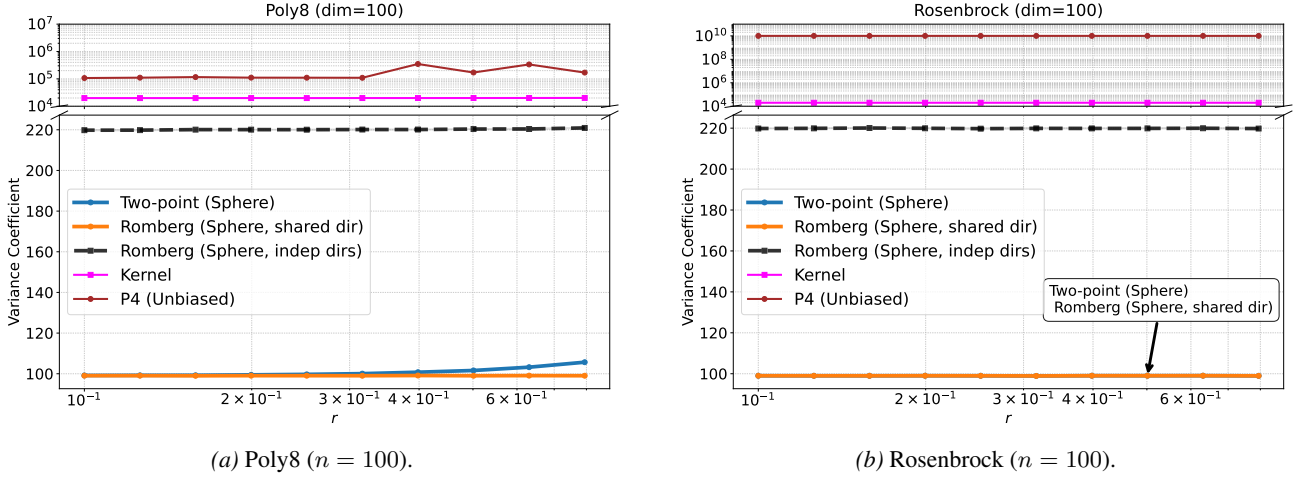
Setup. We use the same number of function evaluations per iteration as in Section 5.2. Specifically, we set the numbers of sampled directions to $m = 2$ for Romberg-ZOGE ($R = 2$), $m = 6$ for Two-point and Kernel, and $m = 3$ for P4. Additional details are provided in Section F.4.

Results. Figure 4 shows that Romberg-ZOGE consistently attains higher values of $f(\mathbf{x})$ than the Two-point baseline under the same number of function evaluations per iteration. In contrast, Kernel and P4 exhibit substantially slower progress, with Kernel remaining nearly flat. Overall, these results indicate that higher-order bias reduction can yield substantial gains under realistic black-box tuning objectives in wireless network optimization.

5.4. Large Language Model Optimization

Finally, we evaluate the proposed estimators in a stochastic-oracle regime by black-box fine-tuning a pretrained large language model (LLM), where each function evaluation is subject to both minibatch noise and sampling noise. Zeroth-order optimization is attractive in this setting due to its memory efficiency, since it avoids storing and backpropagating full gradients (Malladi et al., 2023; Zhang et al., 2024; Yu et al., 2025).

Setup. We fine-tune OPT-1.3B (Zhang et al., 2022) on the Stanford Sentiment Treebank (SST-2) dataset (Socher et al.,


 Figure 1. Bias versus smoothing radius r under $\text{Unif}(\mathbb{S}^{n-1})$.

 Figure 2. Directional variance coefficient $V(r)$ under $\text{Unif}(\mathbb{S}^{n-1})$. Romberg (indep dirs) samples independent directions across radii.

2013) using black-box prompt fine-tuning. Specifically, we optimize a trainable soft prompt of length 10 tokens (i.e., 10×2048 parameters) while keeping the backbone model frozen. To keep the number of function evaluations per iteration the same across methods, we set the numbers of sampled directions to $m = 2$ for Romberg-ZOGE ($R = 2$), $m = 6$ for Two-point and Kernel, and $m = 3$ for P4. Full experimental details are provided in Section F.5.

Results. As shown in Figure 5, Romberg-ZOGE with spherical directions decreases the loss fastest and reaches the lowest value. Romberg-ZOGE with Gaussian directions and P4 follow, with the Gaussian variant converging slightly faster while both attain similar final losses. Two-point is slower, and Kernel exhibits strong oscillations that can even increase the loss, indicating unstable updates in this stochastic regime.

Further quantitative results are presented in Table 1. Un-

der the same total function-evaluation budget per iteration ($B = 12$), Romberg-ZOGE with spherical directions achieves the lowest training loss and the highest validation and test accuracy, indicating that the optimization gains translate into better downstream performance.

 Table 1. Zeroth-order prompt tuning on OPT-1.3B / SST-2 under the same total function-evaluation budget per iteration ($B = 12$).

Method	Train loss ↓	Val. acc. ↑	Test acc. ↑
Romberg (Sphere)	0.262	0.916	0.912
Romberg (Gaussian)	0.323	0.886	0.873
Two-point (Sphere)	0.531	0.828	0.787
Two-point (Gaussian)	0.683	0.600	0.570
P4	0.363	0.890	0.877
Kernel	0.875	0.512	0.552

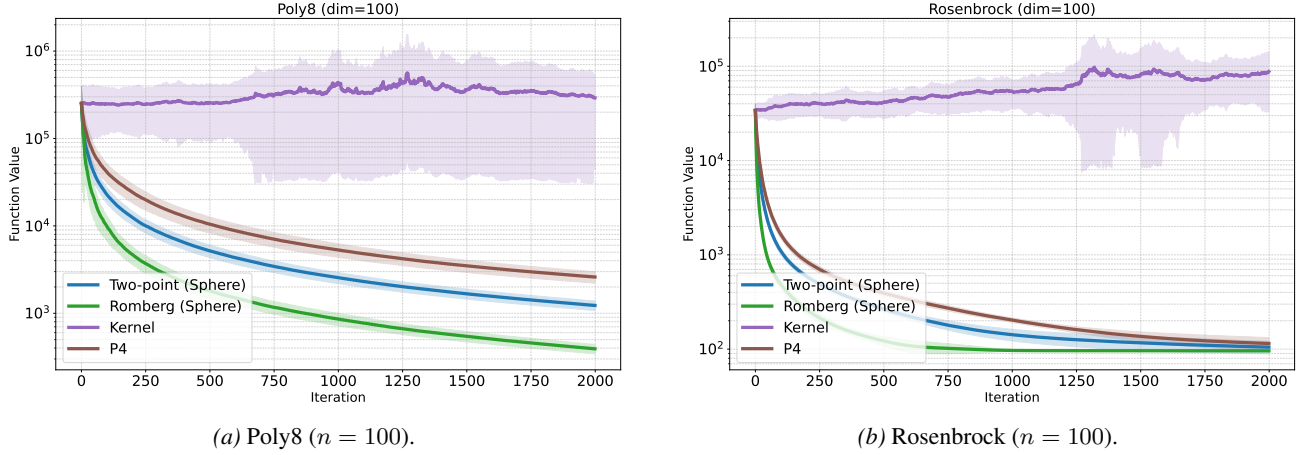


Figure 3. ZO-SGD under $\text{Unif}(\mathbb{S}^{n-1})$ with the same number of function evaluations per iteration.

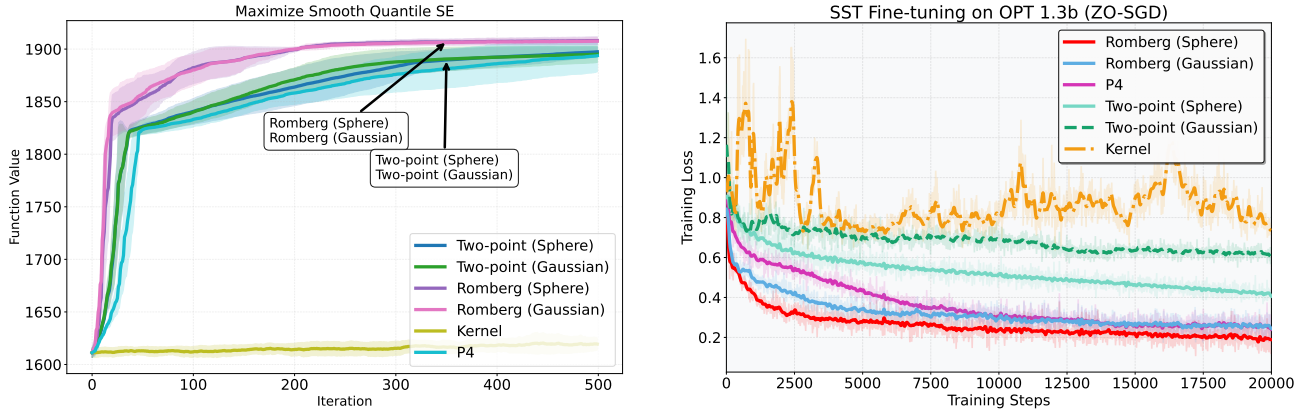


Figure 4. Wireless optimization with a smooth-quantile SE objective.

Figure 5. Black-box fine-tuning loss on OPT-1.3B (SST-2).

6. Conclusion

We proposed Romberg-ZOGE, a Romberg-extrapolated zeroth-order gradient estimator that improves truncation bias to high order while preserving the leading directional variance constant of the standard two-point estimator when the number of function evaluations per gradient estimate is fixed. The central mechanism is to reuse a single perturbation direction across a geometric sequence of radii, which enables pathwise cancellation of the dominant directional randomness. Our theory characterizes the resulting bias and variance in the deterministic setting, and it further quantifies how extrapolation amplifies oracle noise under stochastic function evaluations. Experiments on synthetic objectives, simulator-based wireless network optimization, and black-box LLM prompt tuning show faster and more stable optimization in practice. Overall, these results indicate that extrapolation can deliver higher-order accuracy for zeroth-order gradient estimation without inflating the leading directional variance.

Limitations. Romberg-ZOGE trades higher-order bias reduction for increased noise sensitivity under independent noisy function-value oracles, since larger R increases the κ_R/r^2 term. Hence, R should be chosen according to the oracle noise level. Our theory assumes local high-order smoothness around the queried points, rather than global smoothness of neural-network objectives.

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Impact Statement

This paper aims to advance zeroth-order optimization and its applications in machine learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Akhavan, A., Chzhen, E., Pontil, M., and Tsybakov, A. B. Gradient-free optimization of highly smooth functions: improved analysis and a new algorithm. *Journal of Machine Learning Research*, 25(370):1–50, 2024.
- Bach, F. and Perchet, V. Highly-smooth zero-th order online optimization. In *Conference on Learning Theory*, pp. 257–283. PMLR, 2016.
- Cao, Q., Pun, M.-O., and Chen, Y. Deep learning in network-level performance prediction using cross-layer information. *IEEE Transactions on Network Science and Engineering*, 9(4):2364–2377, 2022. doi: 10.1109/TNSE.2022.3163274.
- Chen, A., Zhang, Y., Jia, J., Diffenderfer, J., Parasiris, K., Liu, J., Zhang, Y., Zhang, Z., Kailkhura, B., and Liu, S. Deepzero: Scaling up zeroth-order optimization for deep model training. In *ICLR*, 2024.
- Chen, G. Unbiased gradient simulation for zeroth-order optimization. In *2020 Winter Simulation Conference (WSC)*, pp. 2947–2959. IEEE, 2020.
- Duchi, J. C., Jordan, M. I., Wainwright, M. J., and Wibisono, A. Optimal rates for zero-order convex optimization: The power of two function evaluations. *IEEE Transactions on Information Theory*, 61(5):2788–2806, 2015.
- Kozak, D., Molinari, C., Rosasco, L., Tenorio, L., and Villa, S. Zeroth-order optimization with orthogonal random directions. *Mathematical Programming*, 199(1-2):1179–1219, 2023.
- Li, Z., Chen, P.-Y., Liu, S., Lu, S., and Xu, Y. Zeroth-order optimization for composite problems with functional constraints. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pp. 7453–7461, 2022.
- Liu, S., Chen, J., Chen, P.-Y., and Hero, A. Zeroth-order online alternating direction method of multipliers: Convergence analysis and applications. In *International Conference on Artificial Intelligence and Statistics*, pp. 288–297. PMLR, 2018a.
- Liu, S., Kailkhura, B., Chen, P.-Y., Ting, P., Chang, S., and Amini, L. Zeroth-order stochastic variance reduction for nonconvex optimization. *Advances in Neural Information Processing Systems*, 31, 2018b.
- Liu, S., Chen, P.-Y., Kailkhura, B., Zhang, G., Hero III, A. O., and Varshney, P. K. A primer on zeroth-order optimization in signal processing and machine learning: Principals, recent advances, and applications. *IEEE Signal Processing Magazine*, 37(5):43–54, 2020.
- Ma, S. and Huang, H. Revisiting zeroth-order optimization: Minimum-variance two-point estimators and directionally aligned perturbations. In *The Thirteenth International Conference on Learning Representations*, 2025a.
- Ma, S. and Huang, H. On the optimal construction of unbiased gradient estimators for zeroth-order optimization. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025b.
- Malladi, S., Gao, T., Nichani, E., Damian, A., Lee, J. D., Chen, D., and Arora, S. Fine-tuning language models with just forward passes. *Advances in Neural Information Processing Systems*, 36:53038–53075, 2023.
- Nesterov, Y. and Spokoiny, V. Random gradient-free minimization of convex functions. *Foundations of Computational Mathematics*, 17:527–566, 2017.
- Socher, R., Perelygin, A., Wu, J., Chuang, J., Manning, C. D., Ng, A. Y., and Potts, C. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 conference on empirical methods in natural language processing*, pp. 1631–1642, 2013.
- Spall, J. C. Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. *IEEE transactions on automatic control*, 37(3):332–341, 1992.
- Yu, Z., Zhou, P., Wang, S., Li, J., Tian, M., and Huang, H. Zeroth-order fine-tuning of llms in random subspaces. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 4475–4485, 2025.
- Zalkind, D. S., Dall’Anese, E., and Pao, L. Y. Automatic controller tuning using a zeroth-order optimization algorithm. *Wind Energy Science*, 5(4):1579–1600, 2020.
- Zhang, S., Roller, S., Goyal, N., Artetxe, M., Chen, M., Chen, S., Dewan, C., Diab, M., Li, X., Lin, X. V., et al. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*, 2022.
- Zhang, S., Xue, Y., Tang, Z., Wang, H., Shen, C., Shi, Q., and Chang, T.-H. Robust network optimization by deep generative models and stochastic optimization. *IEEE Transactions on Wireless Communications*, 24(7):6069–6084, 2025. doi: 10.1109/TWC.2025.3551316.

Zhang, Y., Li, P., Hong, J., Li, J., Zhang, Y., Zheng, W.,
Chen, P.-Y., Lee, J. D., Yin, W., Hong, M., Wang, Z.,
Liu, S., and Chen, T. Revisiting zeroth-order optimization for memory-efficient llm fine-tuning: a benchmark.
In *Proceedings of the 41st International Conference on Machine Learning*, 2024.

A. Proof of Theorem 4.1

We first state a simple moment bound for the two direction distributions considered in this paper.

Lemma A.1. *Let $\mathbf{z} \sim \text{Unif}(\mathbb{S}^{n-1})$ or $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$. Then for any integer $p \geq 0$, there exists a constant $C_p < \infty$ depending only on p (same C_p for all dimensions n) such that*

$$\mathbb{E}_{\mathbf{z}} [\|\mathbf{z}\|^p] \leq C_p \quad \text{for all } n \geq 1.$$

In particular, one may take $C_p = 1$ for $\text{Unif}(\mathbb{S}^{n-1})$, and $C_p = (1+p)^{p/2}$ for $\mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$.

Proof. For $\mathbf{z} \sim \text{Unif}(\mathbb{S}^{n-1})$, we have $\|\mathbf{z}\| = 1$ almost surely, hence $\mathbb{E}_{\mathbf{z}} \|\mathbf{z}\|^p = 1$.

For $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$, write $\mathbf{z} = n^{-1/2} \mathbf{g}$ with $\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n)$. Then $\|\mathbf{g}\|^2 \sim \chi_n^2$ and

$$\mathbb{E}_{\mathbf{z}} \|\mathbf{g}\|^p = 2^{p/2} \frac{\Gamma((n+p)/2)}{\Gamma(n/2)}.$$

Using the standard bound $\Gamma(x+a)/\Gamma(x) \leq x^a$ for $x > 0$ and $a \geq 0$ with $x = n/2$ and $a = p/2$, we obtain

$$\mathbb{E}_{\mathbf{z}} \|\mathbf{z}\|^p = n^{-p/2} \mathbb{E}_{\mathbf{g}} \|\mathbf{g}\|^p \leq n^{-p/2} 2^{p/2} \left(\frac{n}{2}\right)^{p/2} \left(1 + \frac{p}{n}\right)^{p/2} = \left(1 + \frac{p}{n}\right)^{p/2} \leq (1+p)^{p/2},$$

which holds uniformly over all $n \geq 1$. □

We now prove Theorem 4.1.

Proof. Recall the Romberg-ZOGE estimator in (8):

$$\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z}) = \sum_{k=0}^R w_k^{(R)} \mathbf{g}_{2^k}(\mathbf{x}; r_k, \mathbf{z}), \quad \mathbf{g}_{2^k}(\mathbf{x}; r_k, \mathbf{z}) = \frac{n}{2r_k} (f(\mathbf{x} + r_k \mathbf{z}) - f(\mathbf{x} - r_k \mathbf{z})) \mathbf{z},$$

where $r_k = r/2^k$.

Fix $\mathbf{x}$ and a direction $\mathbf{z}$, and define the univariate function $\phi(t) \triangleq f(\mathbf{x} + t\mathbf{z})$. By Taylor's theorem with integral remainder,

$$\phi(t) = \sum_{\ell=0}^{2R+1} \frac{t^\ell}{\ell!} \phi^{(\ell)}(0) + \frac{t^{2R+2}}{(2R+1)!} \int_0^1 (1-u)^{2R+1} \phi^{(2R+2)}(ut) du. \quad (22)$$

Applying (22) to $t = \pm r_k$, subtracting, and dividing by $2r_k$ yields

$$\frac{\phi(r_k) - \phi(-r_k)}{2r_k} = \sum_{j=0}^R \frac{r_k^{2j}}{(2j+1)!} \phi^{(2j+1)}(0) + \tilde{R}_k(\mathbf{x}, \mathbf{z}), \quad (23)$$

where the remainder is

$$\tilde{R}_k(\mathbf{x}, \mathbf{z}) = \frac{r_k^{2R+1}}{2(2R+1)!} \int_0^1 (1-u)^{2R+1} (\phi^{(2R+2)}(ur_k) - \phi^{(2R+2)}(-ur_k)) du. \quad (24)$$

We next bound $\tilde{R}_k(\mathbf{x}, \mathbf{z})$ using the smoothness assumption. Note that $\phi^{(2R+2)}(t) = \nabla^{2R+2} f(\mathbf{x} + t\mathbf{z})[\mathbf{z}, \dots, \mathbf{z}]$, hence

$$\begin{aligned} \left| \phi^{(2R+2)}(ur_k) - \phi^{(2R+2)}(-ur_k) \right| &\leq \left\| \nabla^{2R+2} f(\mathbf{x} + ur_k \mathbf{z}) - \nabla^{2R+2} f(\mathbf{x} - ur_k \mathbf{z}) \right\|_{\text{op}} \|\mathbf{z}\|^{2R+2} \\ &\leq L_{2R+2} \|(\mathbf{x} + ur_k \mathbf{z}) - (\mathbf{x} - ur_k \mathbf{z})\| \|\mathbf{z}\|^{2R+2} \\ &= 2ur_k L_{2R+2} \|\mathbf{z}\|^{2R+3}, \end{aligned}$$

where the second inequality uses Assumption 2.1 with $i = 2R + 2$. Plugging this bound into (24) gives

$$\begin{aligned}
 |\tilde{R}_k(\mathbf{x}, \mathbf{z})| &\leq \frac{r_k^{2R+1}}{2(2R+1)!} \int_0^1 (1-u)^{2R+1} \cdot 2uL_{2R+2} r_k \|\mathbf{z}\|^{2R+3} du \\
 &= \frac{L_{2R+2}}{(2R+1)!} r_k^{2R+2} \|\mathbf{z}\|^{2R+3} \int_0^1 u(1-u)^{2R+1} du \\
 &= \frac{L_{2R+2}}{(2R+1)!} r_k^{2R+2} \|\mathbf{z}\|^{2R+3} \cdot \frac{1}{(2R+2)(2R+3)} \\
 &\leq \frac{L_{2R+2}}{(2R+3)!} r_k^{2R+2} \|\mathbf{z}\|^{2R+3}.
 \end{aligned} \tag{25}$$

Multiplying (23) by $n\mathbf{z}$ yields

$$\mathbf{g}_{2p}(\mathbf{x}; r_k, \mathbf{z}) = n \left(\sum_{j=0}^R \frac{r_k^{2j}}{(2j+1)!} \phi^{(2j+1)}(0) + \tilde{R}_k(\mathbf{x}, \mathbf{z}) \right) \mathbf{z}. \tag{26}$$

Taking expectation over $\mathbf{z}$ and using isotropy $\mathbb{E}_{\mathbf{z}}[\mathbf{z}\mathbf{z}^\top] = \mathbf{I}_n/n$ gives

$$n\mathbb{E}_{\mathbf{z}}[\phi'(0)\mathbf{z}] = n\mathbb{E}_{\mathbf{z}}[\langle \nabla f(\mathbf{x}), \mathbf{z} \rangle \mathbf{z}] = \nabla f(\mathbf{x}).$$

For $j \geq 1$, define

$$\mathbf{c}_{2j}(\mathbf{x}) \triangleq \frac{n}{(2j+1)!} \mathbb{E}_{\mathbf{z}}[\phi^{(2j+1)}(0) \mathbf{z}] = \frac{n}{(2j+1)!} \mathbb{E}_{\mathbf{z}}[\nabla^{(2j+1)} f(\mathbf{x})[\mathbf{z}, \dots, \mathbf{z}]].$$

Then from (26),

$$\mathbb{E}_{\mathbf{z}}[\mathbf{g}_{2p}(\mathbf{x}; r_k, \mathbf{z})] = \nabla f(\mathbf{x}) + \sum_{j=1}^R \mathbf{c}_{2j}(\mathbf{x}) r_k^{2j} + n\mathbb{E}_{\mathbf{z}}[\tilde{R}_k(\mathbf{x}, \mathbf{z})\mathbf{z}]. \tag{27}$$

We now bound the remainder term. By (25) and Cauchy-Schwarz,

$$\begin{aligned}
 \|n\mathbb{E}_{\mathbf{z}}[\tilde{R}_k(\mathbf{x}, \mathbf{z})\mathbf{z}]\| &\leq n \mathbb{E}_{\mathbf{z}}[\|\tilde{R}_k(\mathbf{x}, \mathbf{z})\| \|\mathbf{z}\|] \\
 &\leq \frac{nL_{2R+2}}{(2R+3)!} r_k^{2R+2} \mathbb{E}_{\mathbf{z}}[\|\mathbf{z}\|^{2R+4}] \\
 &\leq \frac{nL_{2R+2}}{(2R+3)!} r_k^{2R+2} C_{2R+4},
 \end{aligned} \tag{28}$$

where C_{2R+4} is from Lemma A.1.

Substituting $r_k = r/2^k$ into (27) yields

$$\mathbb{E}_{\mathbf{z}}[\mathbf{g}_{2p}(\mathbf{x}; r_k, \mathbf{z})] = \nabla f(\mathbf{x}) + \sum_{j=1}^R \mathbf{c}_{2j}(\mathbf{x}) r^{2j} 4^{-jk} + \mathbf{e}_k(\mathbf{x}; r), \tag{29}$$

where the error term satisfies $\|\mathbf{e}_k(\mathbf{x}; r)\| \leq \frac{nL_{2R+2}}{(2R+3)!} C_{2R+4} r^{2R+2}$.

Now apply the Romberg combination. By linearity and (29),

$$\mathbb{E}_{\mathbf{z}}[\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z})] = \left(\sum_{k=0}^R w_k^{(R)} \right) \nabla f(\mathbf{x}) + \sum_{j=1}^R \mathbf{c}_{2j}(\mathbf{x}) r^{2j} \left(\sum_{k=0}^R w_k^{(R)} 4^{-jk} \right) + \sum_{k=0}^R w_k^{(R)} \mathbf{e}_k(\mathbf{x}; r). \tag{30}$$

The weights satisfy $\sum_{k=0}^R w_k^{(R)} = 1$ and $\sum_{k=0}^R w_k^{(R)} 4^{-jk} = 0$ for $j = 1, \dots, R$, so the first two terms in (30) reduce to $\nabla f(\mathbf{x})$.

For the last term, using (28) and $r_k = r/2^k \leq r$,

$$\begin{aligned} \left\| \sum_{k=0}^R w_k^{(R)} e_k(\mathbf{x}; r) \right\| &\leq \sum_{k=0}^R |w_k^{(R)}| \|e_k(\mathbf{x}; r)\| \\ &\leq \frac{nL_{2R+2}}{(2R+3)!} C_{2R+4} r^{2R+2} \sum_{k=0}^R |w_k^{(R)}|. \end{aligned}$$

Define

$$c_b^{(R)} \triangleq \frac{n}{(2R+3)!} C_{2R+4} \sum_{k=0}^R |w_k^{(R)}| L_{2R+2}.$$

We conclude that for all $\mathbf{x} \in \mathbb{R}^n$ and $r > 0$,

$$\left\| \mathbb{E}_{\mathbf{z}} \left[\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z}) \right] - \nabla f(\mathbf{x}) \right\| \leq c_b^{(R)} r^{2R+2},$$

which is exactly (13). □

B. Proof of Theorem 4.2

Proof. Fix any $\mathbf{x} \in \mathbb{R}^n$ and $r \in (0, 1]$. Recall $r_k \triangleq r/2^k$ and

$$\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z}) = \sum_{k=0}^R w_k^{(R)} \mathbf{g}_{2p}(\mathbf{x}; r_k, \mathbf{z}), \quad \mathbf{g}_{2p}(\mathbf{x}; r_k, \mathbf{z}) = \frac{n}{2r_k} (f(\mathbf{x} + r_k \mathbf{z}) - f(\mathbf{x} - r_k \mathbf{z})) \mathbf{z}.$$

Fix a direction $\mathbf{z}$ and define the univariate function $\phi(t) \triangleq f(\mathbf{x} + t\mathbf{z})$. By Taylor's theorem with integral remainder (as in Section A),

$$\phi(t) = \sum_{\ell=0}^{2R+1} \frac{t^\ell}{\ell!} \phi^{(\ell)}(0) + \frac{t^{2R+2}}{(2R+1)!} \int_0^1 (1-u)^{2R+1} \phi^{(2R+2)}(ut) du. \quad (31)$$

Applying (31) to $t = \pm r_k$, subtracting, and dividing by $2r_k$ yields

$$\frac{\phi(r_k) - \phi(-r_k)}{2r_k} = \sum_{j=0}^R \frac{r_k^{2j}}{(2j+1)!} \phi^{(2j+1)}(0) + \tilde{R}_k(\mathbf{x}, \mathbf{z}), \quad (32)$$

where

$$\tilde{R}_k(\mathbf{x}, \mathbf{z}) = \frac{r_k^{2R+1}}{2(2R+1)!} \int_0^1 (1-u)^{2R+1} \left(\phi^{(2R+2)}(ur_k) - \phi^{(2R+2)}(-ur_k) \right) du. \quad (33)$$

Note that $\phi^{(2R+2)}(t) = \nabla^{2R+2} f(\mathbf{x} + t\mathbf{z})[\mathbf{z}, \dots, \mathbf{z}]$. Using Assumption 2.1 with $\beta = 2R+2$ (i.e., Lipschitzness of $\nabla^{2R+2} f$), we have for all $u \in [0, 1]$,

$$\begin{aligned} \left| \phi^{(2R+2)}(ur_k) - \phi^{(2R+2)}(-ur_k) \right| &= \left| \left(\nabla^{2R+2} f(\mathbf{x} + ur_k \mathbf{z}) - \nabla^{2R+2} f(\mathbf{x} - ur_k \mathbf{z}) \right) [\mathbf{z}, \dots, \mathbf{z}] \right| \\ &\leq \left\| \nabla^{2R+2} f(\mathbf{x} + ur_k \mathbf{z}) - \nabla^{2R+2} f(\mathbf{x} - ur_k \mathbf{z}) \right\|_{\text{op}} \|\mathbf{z}\|^{2R+2} \\ &\leq L_{2R+2} \left\| (\mathbf{x} + ur_k \mathbf{z}) - (\mathbf{x} - ur_k \mathbf{z}) \right\| \|\mathbf{z}\|^{2R+2} \\ &= 2uL_{2R+2} r_k \|\mathbf{z}\|^{2R+3}. \end{aligned}$$

Plugging this into (33) gives

$$\begin{aligned}
 |\tilde{R}_k(\mathbf{x}, \mathbf{z})| &\leq \frac{r_k^{2R+1}}{2(2R+1)!} \int_0^1 (1-u)^{2R+1} \cdot 2uL_{2R+2} r_k \|\mathbf{z}\|^{2R+3} du \\
 &= \frac{L_{2R+2}}{(2R+1)!} r_k^{2R+2} \|\mathbf{z}\|^{2R+3} \int_0^1 u(1-u)^{2R+1} du \\
 &= \frac{L_{2R+2}}{(2R+1)!} r_k^{2R+2} \|\mathbf{z}\|^{2R+3} \cdot \frac{1}{(2R+2)(2R+3)} \\
 &\leq \frac{L_{2R+2}}{(2R+3)!} r_k^{2R+2} \|\mathbf{z}\|^{2R+3}.
 \end{aligned} \tag{34}$$

Multiplying (32) by $n\mathbf{z}$ yields the *pathwise* expansion

$$g_{2p}(\mathbf{x}; r_k, \mathbf{z}) = g_M(\mathbf{x}, \mathbf{z}) + \sum_{j=1}^R r_k^{2j} \mathbf{T}_{2j}(\mathbf{x}, \mathbf{z}) + \mathbf{Rem}_k(\mathbf{x}; r_k, \mathbf{z}), \tag{35}$$

where

$$g_M(\mathbf{x}, \mathbf{z}) \triangleq n\phi'(0)\mathbf{z} = n\langle \nabla f(\mathbf{x}), \mathbf{z} \rangle \mathbf{z}, \quad \mathbf{T}_{2j}(\mathbf{x}, \mathbf{z}) \triangleq \frac{n}{(2j+1)!} \phi^{(2j+1)}(0) \mathbf{z}, \quad \mathbf{Rem}_k(\mathbf{x}; r_k, \mathbf{z}) \triangleq n \tilde{R}_k(\mathbf{x}, \mathbf{z}) \mathbf{z}.$$

By (34), the remainder satisfies the pointwise bound

$$\|\mathbf{Rem}_k(\mathbf{x}; r_k, \mathbf{z})\| \leq \frac{nL_{2R+2}}{(2R+3)!} r_k^{2R+2} \|\mathbf{z}\|^{2R+4}. \tag{36}$$

Summing (35) with weights $\{w_k^{(R)}\}_{k=0}^R$ gives

$$\begin{aligned}
 g^{[R]}(\mathbf{x}; r, \mathbf{z}) &= \sum_{k=0}^R w_k^{(R)} g_{2p}(\mathbf{x}; r_k, \mathbf{z}) \\
 &= \left(\sum_{k=0}^R w_k^{(R)} \right) g_M(\mathbf{x}, \mathbf{z}) + \sum_{j=1}^R r^{2j} \mathbf{T}_{2j}(\mathbf{x}, \mathbf{z}) \left(\sum_{k=0}^R w_k^{(R)} 4^{-jk} \right) + \sum_{k=0}^R w_k^{(R)} \mathbf{Rem}_k(\mathbf{x}; r_k, \mathbf{z}).
 \end{aligned} \tag{37}$$

Using the Romberg conditions (11),

$$\sum_{k=0}^R w_k^{(R)} = 1, \quad \sum_{k=0}^R w_k^{(R)} 4^{-jk} = 0, \quad j = 1, \dots, R,$$

the truncation terms cancel *pathwise* in (37), yielding the exact decomposition

$$g^{[R]}(\mathbf{x}; r, \mathbf{z}) = g_M(\mathbf{x}, \mathbf{z}) + \Xi(\mathbf{x}; r, \mathbf{z}), \quad \Xi(\mathbf{x}; r, \mathbf{z}) \triangleq \sum_{k=0}^R w_k^{(R)} \mathbf{Rem}_k(\mathbf{x}; r_k, \mathbf{z}). \tag{38}$$

By (36) and $r_k = r/2^k \leq r$,

$$\begin{aligned}
 \|\Xi(\mathbf{x}; r, \mathbf{z})\| &\leq \sum_{k=0}^R |w_k^{(R)}| \|\mathbf{Rem}_k(\mathbf{x}; r_k, \mathbf{z})\| \\
 &\leq \frac{nL_{2R+2}}{(2R+3)!} \|\mathbf{z}\|^{2R+4} \sum_{k=0}^R |w_k^{(R)}| r_k^{2R+2} \\
 &= \frac{nL_{2R+2}}{(2R+3)!} r^{2R+2} \|\mathbf{z}\|^{2R+4} \sum_{k=0}^R |w_k^{(R)}| 2^{-(2R+2)k} \leq \frac{nL_{2R+2}}{(2R+3)!} r^{2R+2} \|\mathbf{z}\|^{2R+4} \sum_{k=0}^R |w_k^{(R)}|.
 \end{aligned} \tag{39}$$

By Lemma A.1 (with $p = 2R + 4$ and $p = 4R + 8$), from (39) we obtain

$$\mathbb{E}_{\mathbf{z}} \|\Xi(\mathbf{x}; r, \mathbf{z})\| \leq \frac{nL_{2R+2}}{(2R+3)!} \left(\sum_{k=0}^R |w_k^{(R)}| \right) r^{2R+2} \mathbb{E}_{\mathbf{z}} \|\mathbf{z}\|^{2R+4} \triangleq a_R r^{2R+2}, \quad (40)$$

$$\mathbb{E}_{\mathbf{z}} \|\Xi(\mathbf{x}; r, \mathbf{z})\|^2 \leq \left(\frac{nL_{2R+2}}{(2R+3)!} \right)^2 \left(\sum_{k=0}^R |w_k^{(R)}| \right)^2 r^{4R+4} \mathbb{E}_{\mathbf{z}} \|\mathbf{z}\|^{4R+8} \triangleq b_R r^{4R+4}. \quad (41)$$

In particular, $\text{Var}_{\mathbf{z}}(\Xi) \leq \mathbb{E}_{\mathbf{z}} \|\Xi\|^2 \leq b_R r^{4R+4}$.

Using (38) and the identity $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$ with $\text{Cov}(X, Y) \triangleq \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$, we have

$$\text{Var}_{\mathbf{z}}(\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z})) = \text{Var}_{\mathbf{z}}(\mathbf{g}_M(\mathbf{x}, \mathbf{z})) + \text{Var}_{\mathbf{z}}(\Xi(\mathbf{x}; r, \mathbf{z})) + 2 \text{Cov}_{\mathbf{z}}(\mathbf{g}_M(\mathbf{x}, \mathbf{z}), \Xi(\mathbf{x}; r, \mathbf{z})). \quad (42)$$

For isotropic symmetric directions (Lemma 2.1),

$$\text{Var}_{\mathbf{z}}(\mathbf{g}_M(\mathbf{x}, \mathbf{z})) = V \|\nabla f(\mathbf{x})\|^2, \quad (43)$$

where $V = n - 1$ if $\mathbf{z} \sim \text{Unif}(\mathbb{S}^{n-1})$ and $V = n + 1$ if $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$.

Moreover, by (41), $\text{Var}_{\mathbf{z}}(\Xi) \leq b_R r^{4R+4}$.

Finally, by Cauchy–Schwarz,

$$|\text{Cov}_{\mathbf{z}}(\mathbf{g}_M, \Xi)| \leq \sqrt{\text{Var}_{\mathbf{z}}(\mathbf{g}_M) \text{Var}_{\mathbf{z}}(\Xi)} \leq \sqrt{V} \|\nabla f(\mathbf{x})\| \sqrt{\text{Var}_{\mathbf{z}}(\Xi)} \leq \sqrt{V} \|\nabla f(\mathbf{x})\| \sqrt{\mathbb{E}_{\mathbf{z}} \|\Xi\|^2}.$$

Thus, using (41),

$$2|\text{Cov}_{\mathbf{z}}(\mathbf{g}_M, \Xi)| \leq 2\sqrt{V} \|\nabla f(\mathbf{x})\| \cdot \sqrt{b_R} r^{2R+2}. \quad (44)$$

Substituting (43), $\text{Var}_{\mathbf{z}}(\Xi) \leq b_R r^{4R+4}$, and (44) into (42) yields

$$\text{Var}_{\mathbf{z}}(\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z})) \leq V \|\nabla f(\mathbf{x})\|^2 + \underbrace{2\sqrt{Vb_R}}_{=: c_{v,1}^{(R)}} \|\nabla f(\mathbf{x})\| r^{2R+2} + \underbrace{b_R}_{=: c_{v,2}^{(R)}} r^{4R+4},$$

which is exactly (14). □

C. Proof of Theorem 4.3

Proof. We prove the bias and variance statements in turn. Recall that each oracle call returns $\tilde{f}(\mathbf{x}; \xi)$ with $\mathbb{E}_{\xi}[\tilde{f}(\mathbf{x}; \xi)] = f(\mathbf{x})$ and $\text{Var}_{\xi}(\tilde{f}(\mathbf{x}; \xi)) \leq \sigma^2$. Different oracle calls use independent copies of ξ .

1. Bias bound. Fix $\mathbf{z}$ and write the $2(R+1)$ oracle noises in one estimator as $\xi^{[R]} = \{\xi_{k,+}, \xi_{k,-}\}_{k=0}^R$. Using unbiasedness for each query point,

$$\begin{aligned} \mathbb{E}_{\xi^{[R]}}[\tilde{\mathbf{g}}^{[R]}(\mathbf{x}; r, \mathbf{z}) \mid \mathbf{z}] &= \sum_{k=0}^R w_k^{(R)} \frac{n}{2r_k} \left(\mathbb{E}_{\xi_{k,+}}[\tilde{f}(\mathbf{x} + r_k \mathbf{z}; \xi_{k,+})] - \mathbb{E}_{\xi_{k,-}}[\tilde{f}(\mathbf{x} - r_k \mathbf{z}; \xi_{k,-})] \right) \mathbf{z} \\ &= \sum_{k=0}^R w_k^{(R)} \frac{n}{2r_k} \left(f(\mathbf{x} + r_k \mathbf{z}) - f(\mathbf{x} - r_k \mathbf{z}) \right) \mathbf{z} \\ &= \mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z}). \end{aligned}$$

Taking expectation over $\mathbf{z}$ gives $\mathbb{E}_{\mathbf{z}, \xi^{[R]}}[\tilde{\mathbf{g}}^{[R]}(\mathbf{x}; r, \mathbf{z})] = \mathbb{E}_{\mathbf{z}}[\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z})]$, and Theorem 4.1 yields (15).

2. Variance bound. Decompose the estimator into the deterministic part plus a pure-noise term. For each $k = 0, \dots, R$, let $r_k = r/2^k$ and define

$$\tilde{\Delta}_k \triangleq \tilde{f}(\mathbf{x} + r_k \mathbf{z}; \boldsymbol{\xi}_{k,+}) - \tilde{f}(\mathbf{x} - r_k \mathbf{z}; \boldsymbol{\xi}_{k,-}) = \Delta_k + \delta_k,$$

where $\Delta_k = f(\mathbf{x} + r_k \mathbf{z}) - f(\mathbf{x} - r_k \mathbf{z})$ and $\delta_k = (\tilde{f}(\mathbf{x} + r_k \mathbf{z}; \boldsymbol{\xi}_{k,+}) - f(\mathbf{x} + r_k \mathbf{z})) - (\tilde{f}(\mathbf{x} - r_k \mathbf{z}; \boldsymbol{\xi}_{k,-}) - f(\mathbf{x} - r_k \mathbf{z}))$. Then

$$\tilde{\mathbf{g}}^{[R]}(\mathbf{x}; r, \mathbf{z}) = \sum_{k=0}^R w_k^{(R)} \frac{n}{2r_k} \tilde{\Delta}_k \mathbf{z} = \mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z}) + N(\mathbf{x}; r, \mathbf{z}),$$

with

$$N(\mathbf{x}; r, \mathbf{z}) \triangleq \sum_{k=0}^R w_k^{(R)} \frac{n}{2r_k} \delta_k \mathbf{z}.$$

Using the law of total variance with respect to $\mathbf{z}$,

$$\text{Var}_{\mathbf{z}, \boldsymbol{\xi}^{[R]}}(\tilde{\mathbf{g}}^{[R]}(\mathbf{x}; r, \mathbf{z})) = \text{Var}_{\mathbf{z}}(\mathbb{E}_{\boldsymbol{\xi}^{[R]}}[\tilde{\mathbf{g}}^{[R]}(\mathbf{x}; r, \mathbf{z}) \mid \mathbf{z}]) + \mathbb{E}_{\mathbf{z}}[\text{Var}_{\boldsymbol{\xi}^{[R]}}(\tilde{\mathbf{g}}^{[R]}(\mathbf{x}; r, \mathbf{z}) \mid \mathbf{z})]. \quad (45)$$

The first term equals the deterministic variance and is bounded by Theorem 4.2:

$$\text{Var}_{\mathbf{z}}(\mathbb{E}_{\boldsymbol{\xi}^{[R]}}[\tilde{\mathbf{g}}^{[R]}(\mathbf{x}; r, \mathbf{z}) \mid \mathbf{z}]) = \text{Var}_{\mathbf{z}}(\mathbf{g}^{[R]}(\mathbf{x}; r, \mathbf{z})) \leq V \|\nabla f(\mathbf{x})\|^2 + c_{v,1}^{(R)} \|\nabla f(\mathbf{x})\| r^{2R+2} + c_{v,2}^{(R)} r^{4R+4}.$$

Fix $\mathbf{z}$. Since $r_k \neq r_{k'}$ for $k \neq k'$, the points $\mathbf{x} \pm r_k \mathbf{z}$ are distinct as long as $\mathbf{z} \neq \mathbf{0}$. For both direction distributions considered, $\mathbb{P}(\mathbf{z} = \mathbf{0}) = 0$ (indeed $\|\mathbf{z}\| = 1$ a.s. for $\text{Unif}(\mathbb{S}^{n-1})$, and $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$ is non-degenerate). Hence the $2(R+1)$ query points $\{\mathbf{x} \pm r_k \mathbf{z}\}_{k=0}^R$ are pairwise distinct almost surely. By Assumption 4.1 (independence across distinct query points), the oracle noises across different radii are independent conditional on $\mathbf{z}$; in particular, $\{\delta_k\}_{k=0}^R$ are independent given $\mathbf{z}$.

By independence across query points and bounded variance, δ_k are independent across k conditional on $\mathbf{z}$, and

$$\text{Var}_{\boldsymbol{\xi}^{[R]}}(\delta_k \mid \mathbf{z}) \leq \text{Var}_{\boldsymbol{\xi}_{k,+}}(\tilde{f}(\mathbf{x} + r_k \mathbf{z}; \boldsymbol{\xi}_{k,+})) + \text{Var}_{\boldsymbol{\xi}_{k,-}}(\tilde{f}(\mathbf{x} - r_k \mathbf{z}; \boldsymbol{\xi}_{k,-})) \leq 2\sigma^2.$$

Therefore,

$$\begin{aligned} \text{Var}_{\boldsymbol{\xi}^{[R]}}(\tilde{\mathbf{g}}^{[R]}(\mathbf{x}; r, \mathbf{z}) \mid \mathbf{z}) &= \text{Var}_{\boldsymbol{\xi}^{[R]}}(N(\mathbf{x}; r, \mathbf{z}) \mid \mathbf{z}) \\ &= n^2 \|\mathbf{z}\|^2 \sum_{k=0}^R \frac{(w_k^{(R)})^2}{4r_k^2} \text{Var}_{\boldsymbol{\xi}^{[R]}}(\delta_k \mid \mathbf{z}) \\ &\leq \frac{n^2 \sigma^2}{2} \|\mathbf{z}\|^2 \sum_{k=0}^R \frac{(w_k^{(R)})^2}{r_k^2}. \end{aligned}$$

Using $r_k = r/2^k$ gives

$$\sum_{k=0}^R \frac{(w_k^{(R)})^2}{r_k^2} = \frac{1}{r^2} \sum_{k=0}^R (w_k^{(R)})^2 4^k = \frac{\kappa_R}{r^2}.$$

Taking expectation over $\mathbf{z}$, note that $\mathbb{E}\|\mathbf{z}\|^2 = 1$ for both direction distributions: if $\mathbf{z} \sim \text{Unif}(\mathbb{S}^{n-1})$, then $\|\mathbf{z}\| = 1$ a.s.; if $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$, then $\mathbb{E}\|\mathbf{z}\|^2 = \text{tr}(\mathbb{E}[\mathbf{z}\mathbf{z}^\top]) = \text{tr}(\mathbf{I}_n/n) = 1$.

Hence,

$$\mathbb{E}_{\mathbf{z}}[\text{Var}_{\boldsymbol{\xi}^{[R]}}(\tilde{\mathbf{g}}^{[R]}(\mathbf{x}; r, \mathbf{z}) \mid \mathbf{z})] \leq \frac{n^2 \sigma^2}{2} \cdot \mathbb{E}_{\mathbf{z}}\|\mathbf{z}\|^2 \cdot \frac{\kappa_R}{r^2} = \frac{n^2 \sigma^2}{2r^2} \kappa_R,$$

Substituting into (45) gives (16). □

D. Proof of Theorem 4.4

Proof. For each iteration t , the algorithm samples a direction $\mathbf{z}_t \sim \mathcal{D}$ and uses $2(R+1)$ independent oracle noises $\boldsymbol{\xi}_t^{[R]}$ to form $\hat{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t)$. Define the history sigma-algebra

$$\Omega_t \triangleq \sigma(\mathbf{z}_0, \dots, \mathbf{z}_t, \boldsymbol{\xi}_0^{[R]}, \dots, \boldsymbol{\xi}_t^{[R]}), \quad t \geq 0,$$

and write $\mathbb{E}_{\Omega_t}[\cdot]$ for expectation with respect to all randomness in Ω_t . Note that $\mathbf{x}_t$ is Ω_{t-1} -measurable.

Fix $t \in \{0, \dots, T-1\}$. By L_1 -smoothness of f ,

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) - \eta \langle \nabla f(\mathbf{x}_t), \hat{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t) \rangle + \frac{L_1 \eta^2}{2} \|\hat{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t)\|^2. \quad (46)$$

Taking conditional expectation w.r.t. the new randomness $(\mathbf{z}_t, \boldsymbol{\xi}_t^{[R]})$ given Ω_{t-1} (i.e., $\mathbb{E}[\cdot \mid \Omega_{t-1}]$), define the conditional mean and bias

$$\bar{g}_t \triangleq \mathbb{E}[\hat{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t) \mid \Omega_{t-1}], \quad b_t \triangleq \bar{g}_t - \nabla f(\mathbf{x}_t).$$

Then

$$\mathbb{E}[\langle \nabla f(\mathbf{x}_t), \hat{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t) \rangle \mid \Omega_{t-1}] = \|\nabla f(\mathbf{x}_t)\|^2 + \langle \nabla f(\mathbf{x}_t), b_t \rangle, \quad (47)$$

$$\mathbb{E}[\|\hat{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t)\|^2 \mid \Omega_{t-1}] = \|\bar{g}_t\|^2 + \text{Var}(\hat{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t) \mid \Omega_{t-1}). \quad (48)$$

By Theorem 4.3 (or Theorem 4.1 when $\sigma = 0$),

$$\|b_t\| \leq c_b^{(R)} r^{2R+2}, \quad \text{so} \quad \|b_t\|^2 \leq (c_b^{(R)})^2 r^{4R+4}. \quad (49)$$

Moreover, by Theorem 4.3 and Ω_{t-1} -measurability of $\mathbf{x}_t$,

$$\text{Var}(\hat{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t) \mid \Omega_{t-1}) \leq V \|\nabla f(\mathbf{x}_t)\|^2 + c_{v,1}^{(R)} \|\nabla f(\mathbf{x}_t)\| r^{2R+2} + c_{v,2}^{(R)} r^{4R+4} + \frac{n^2 \sigma^2}{2r^2} \kappa_R. \quad (50)$$

Apply Young's inequality: for any $\alpha > 0$,

$$c_{v,1}^{(R)} \|\nabla f(\mathbf{x}_t)\| r^{2R+2} \leq \alpha \|\nabla f(\mathbf{x}_t)\|^2 + \frac{(c_{v,1}^{(R)})^2}{4\alpha} r^{4R+4}.$$

Choose $\alpha = 1$ and define

$$\tilde{c}_{v,2}^{(R)} \triangleq c_{v,2}^{(R)} + \frac{(c_{v,1}^{(R)})^2}{4}.$$

Then (50) implies

$$\text{Var}(\hat{\mathbf{g}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t) \mid \Omega_{t-1}) \leq (V+1) \|\nabla f(\mathbf{x}_t)\|^2 + \tilde{c}_{v,2}^{(R)} r^{4R+4} + \frac{n^2 \sigma^2}{2r^2} \kappa_R. \quad (51)$$

Taking $\mathbb{E}[\cdot \mid \Omega_{t-1}]$ in (46) and using (47)–(48) yields

$$\mathbb{E}[f(\mathbf{x}_{t+1}) \mid \Omega_{t-1}] \leq f(\mathbf{x}_t) - \eta \|\nabla f(\mathbf{x}_t)\|^2 - \eta \langle \nabla f(\mathbf{x}_t), b_t \rangle + \frac{L_1 \eta^2}{2} \|\bar{g}_t\|^2 + \frac{L_1 \eta^2}{2} \text{Var}(\hat{\mathbf{g}}^{[R]} \mid \Omega_{t-1}). \quad (52)$$

Use $\|\bar{g}_t\|^2 = \|\nabla f(\mathbf{x}_t) + b_t\|^2 \leq 2\|\nabla f(\mathbf{x}_t)\|^2 + 2\|b_t\|^2$ and $-\langle \nabla f(\mathbf{x}_t), b_t \rangle \leq \frac{1}{4} \|\nabla f(\mathbf{x}_t)\|^2 + \|b_t\|^2$ to get

$$\mathbb{E}[f(\mathbf{x}_{t+1}) \mid \Omega_{t-1}] \leq f(\mathbf{x}_t) - \eta \left(\frac{3}{4} - L_1 \eta \right) \|\nabla f(\mathbf{x}_t)\|^2 + (\eta + L_1 \eta^2) \|b_t\|^2 + \frac{L_1 \eta^2}{2} \text{Var}(\hat{\mathbf{g}}^{[R]} \mid \Omega_{t-1}). \quad (53)$$

Substitute (51) into (53). With $\eta \leq \frac{1}{8L_1(V+1)}$, we have $\frac{3}{4} - L_1\eta - \frac{L_1\eta}{2}(V+1) \geq \frac{1}{2}$ and $1 + L_1\eta \leq 2$, hence

$$\mathbb{E}[f(\mathbf{x}_{t+1}) \mid \Omega_{t-1}] \leq f(\mathbf{x}_t) - \frac{\eta}{2} \|\nabla f(\mathbf{x}_t)\|^2 + 2\eta \|b_t\|^2 + \frac{L_1\eta^2}{2} \left(\tilde{c}_{v,2}^{(R)} r^{4R+4} + \frac{n^2\sigma^2}{2r^2} \kappa_R \right). \quad (54)$$

Finally plug (49) into (54).

Now take expectation with respect to Ω_{t-1} on both sides of the last inequality. Using the tower property $\mathbb{E}_{\Omega_t}[\cdot] = \mathbb{E}_{\Omega_{t-1}}[\mathbb{E}[\cdot \mid \Omega_{t-1}]]$ and summing over $t = 0, \dots, T-1$, we get

$$\mathbb{E}_{\Omega_{T-1}}[f(\mathbf{x}_T)] \leq f(\mathbf{x}_0) - \frac{\eta}{2} \sum_{t=0}^{T-1} \mathbb{E}_{\Omega_t} \|\nabla f(\mathbf{x}_t)\|^2 + 2\eta T (c_b^{(R)})^2 r^{4R+4} + \frac{L_1\eta^2 T}{2} \left(\tilde{c}_{v,2}^{(R)} r^{4R+4} + \frac{n^2\sigma^2}{2r^2} \kappa_R \right).$$

Since $f(\mathbf{x}_T) \geq f^*$ deterministically, $\mathbb{E}_{\Omega_{T-1}}[f(\mathbf{x}_T)] \geq f^*$, thus

$$\frac{\eta}{2} \sum_{t=0}^{T-1} \mathbb{E}_{\Omega_t} \|\nabla f(\mathbf{x}_t)\|^2 \leq f(\mathbf{x}_0) - f^* + 2\eta T (c_b^{(R)})^2 r^{4R+4} + \frac{L_1\eta^2 T}{2} \left(\tilde{c}_{v,2}^{(R)} r^{4R+4} + \frac{n^2\sigma^2}{2r^2} \kappa_R \right).$$

Divide both sides by $\eta T/2$ to obtain

$$\min_{0 \leq t \leq T-1} \mathbb{E}_{\Omega_t} \|\nabla f(\mathbf{x}_t)\|^2 \leq \frac{2(f(\mathbf{x}_0) - f^*)}{\eta T} + L_1\eta \cdot \frac{n^2\sigma^2}{2r^2} \kappa_R + \left(4(c_b^{(R)})^2 + L_1\eta \tilde{c}_{v,2}^{(R)} \right) r^{4R+4}.$$

Since $\eta \leq \frac{1}{8L_1(V+1)}$, the coefficient of r^{4R+4} is upper bounded by a constant $C_R > 0$ depending only on $n, R, \mathcal{D}, L_1, V$ and the estimator constants, and independent of T, η, r . Absorbing constants yields (19). $\square$

E. Proof of Corollary 4.1

Proof. We start from the bound in Theorem 4.4: for any $T \geq 1$ and any $r \in (0, 1]$,

$$\min_{0 \leq t \leq T-1} \mathbb{E}_{\Omega_{T-1}} \|\nabla f(\mathbf{x}_t)\|^2 \leq \frac{2(f(\mathbf{x}_0) - f^*)}{\eta T} + L_1\eta \cdot \frac{n^2\sigma^2}{2r^2} \kappa_R + C_R r^{4R+4}. \quad (55)$$

For readability, denote the three terms on the right-hand side by

$$(I) \triangleq \frac{2(f(\mathbf{x}_0) - f^*)}{\eta T}, \quad (II) \triangleq L_1\eta \cdot \frac{n^2\sigma^2}{2r^2} \kappa_R, \quad (III) \triangleq C_R r^{4R+4}.$$

It suffices to choose parameters so that each term is at most $\varepsilon^2/3$.

1) Deterministic oracle ($\sigma = 0$). Then $(II) = 0$. Choose any constant stepsize $0 < \eta \leq \frac{1}{8L_1(V+1)}$ and choose $r \leq \left(\frac{\varepsilon^2}{3C_R} \right)^{\frac{1}{4R+4}}$, which ensures $(III) \leq \varepsilon^2/3$. Now pick T large enough so that $(I) \leq \varepsilon^2/3$; since η is a constant, this yields an iteration complexity scaling $T = \mathcal{O}(\varepsilon^{-2})$ after absorbing all ε -independent quantities into the big- $\mathcal{O}$ notation. This proves the deterministic case.

2) Stochastic oracle ($\sigma > 0$). Choose $r \leq \left(\frac{\varepsilon^2}{3C_R} \right)^{\frac{1}{4R+4}}$, so that $(III) \leq \varepsilon^2/3$. Next choose the stepsize

$$\eta \leq \min \left\{ \frac{1}{8L_1(V+1)}, \frac{2\varepsilon^2 r^2}{3L_1 n^2 \sigma^2 \kappa_R} \right\},$$

which ensures $(II) \leq \varepsilon^2/3$. Finally pick T large enough so that $(I) \leq \varepsilon^2/3$, i.e., $T = \mathcal{O}(1/(\eta\varepsilon^2))$.

It remains to express this in terms of ε only. With $r \asymp \varepsilon^{\frac{1}{2R+2}}$, we have $r^2 \asymp \varepsilon^{\frac{1}{R+1}}$, hence the chosen stepsize satisfies $\eta \asymp \varepsilon^{2 + \frac{1}{R+1}}$ up to ε -independent constants. Therefore,

$$T = \mathcal{O}\left(\frac{1}{\eta\varepsilon^2}\right) = \mathcal{O}\left(\varepsilon^{-(4 + \frac{1}{R+1})}\right),$$

after absorbing all ε -independent constants into the big- $\mathcal{O}$ notation. This proves the stochastic case and completes the proof. $\square$

E.1. Shared-Sample Stochastic-Objective Oracle

We provide the proof of the claim stated in Remark 4.5. This result considers a stochastic-objective oracle in which the same sample is reused across all queried points within one Romberg estimator.

Assumption E.1 (Shared-sample stochastic objective). *There exists a random variable ξ and a stochastic objective $\tilde{f}(\cdot; \xi)$ such that*

$$f(\mathbf{x}) = \mathbb{E}_{\xi}[\tilde{f}(\mathbf{x}; \xi)], \quad \mathbb{E}_{\xi}[\nabla \tilde{f}(\mathbf{x}; \xi)] = \nabla f(\mathbf{x}). \quad (56)$$

Moreover, the stochastic gradient has bounded variance:

$$\mathbb{E}_{\xi} \|\nabla \tilde{f}(\mathbf{x}; \xi) - \nabla f(\mathbf{x})\|^2 \leq \sigma_g^2, \quad \forall \mathbf{x} \in \mathbb{R}^n. \quad (57)$$

For almost every ξ , the sample function $\tilde{f}(\cdot; \xi)$ satisfies the same local $(2R + 2)$ -order smoothness condition as in Assumption 2.1, with constants controlled uniformly in expectation.

At iteration t , the shared-sample Romberg estimator is defined as

$$\tilde{\mathbf{g}}_{\text{sh}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t, \xi_t) \triangleq \sum_{k=0}^R w_k^{(R)} \frac{n}{2r_k} \left(\tilde{f}(\mathbf{x}_t + r_k \mathbf{z}_t; \xi_t) - \tilde{f}(\mathbf{x}_t - r_k \mathbf{z}_t; \xi_t) \right) \mathbf{z}_t, \quad (58)$$

where $r_k = r/2^k$. The same sample ξ_t is used for all $k = 0, \dots, R$, while fresh independent samples are drawn across iterations.

Proposition E.1 (ZO-SGD under the shared-sample stochastic objective). *Suppose Assumptions 4.2, 2.1, and E.1 hold. Run ZO-SGD with the estimator (58) and stepsize $0 < \eta \leq c/(L_1(V + 1))$ for a sufficiently small numerical constant $c > 0$. Then, for any $T \geq 1$ and $r \in (0, 1]$,*

$$\begin{aligned} & \min_{0 \leq t \leq T-1} \mathbb{E} \|\nabla f(\mathbf{x}_t)\|^2 \\ & \leq \frac{2(f(\mathbf{x}_0) - f^*)}{\eta T} + \tilde{C}_1 n \eta \sigma_g^2 + \tilde{C}_R r^{4R+4}, \end{aligned} \quad (59)$$

where $\tilde{C}_1$ and $\tilde{C}_R$ are positive constants independent of T, η, r, σ_g , and ε .

Proof. Fix an iterate $\mathbf{x}_t$ and write

$$\tilde{\mathbf{g}}_t = \tilde{\mathbf{g}}_{\text{sh}}^{[R]}(\mathbf{x}_t; r, \mathbf{z}_t, \xi_t).$$

For a fixed sample ξ_t , the estimator (58) is exactly the deterministic Romberg-ZOGE applied to the sample function $\tilde{f}(\cdot; \xi_t)$. Therefore, by the same Romberg bias-cancellation argument used in Theorem 4.1, and then taking expectation over ξ_t , we have

$$\|\mathbb{E}_{\mathbf{z}_t, \xi_t}[\tilde{\mathbf{g}}_t \mid \mathbf{x}_t] - \nabla f(\mathbf{x}_t)\| \leq \tilde{c}_b^{(R)} r^{2R+2}. \quad (60)$$

Next, we bound the second moment. For fixed ξ_t , the leading term of the shared-sample Romberg estimator is

$$n \mathbf{z}_t \mathbf{z}_t^\top \nabla \tilde{f}(\mathbf{x}_t; \xi_t),$$

and the remaining Romberg truncation residual is of order r^{2R+2} . Using the isotropy of $\mathbf{z}_t$, the directional second-moment bound, and Assumption E.1, we obtain

$$\mathbb{E}_{\mathbf{z}_t, \xi_t} [\|\tilde{\mathbf{g}}_t\|^2 \mid \mathbf{x}_t] \leq \tilde{C}_0(V + 1) \|\nabla f(\mathbf{x}_t)\|^2 + \tilde{C}_1 n \sigma_g^2 + \tilde{C}_2^{(R)} r^{4R+4}. \quad (61)$$

The key point is that there is no term of the form $\kappa_R \sigma^2 / r^2$. This is because the same sample ξ_t is reused across all radii, so the stochasticity enters through the sample gradient $\nabla \tilde{f}(\mathbf{x}_t; \xi_t)$ rather than through independent additive noises at the $2(R + 1)$ queried function values.

Since f is L_1 -smooth,

$$\mathbb{E}[f(\mathbf{x}_{t+1}) \mid \mathbf{x}_t] \leq f(\mathbf{x}_t) - \eta \langle \nabla f(\mathbf{x}_t), \mathbb{E}[\tilde{\mathbf{g}}_t \mid \mathbf{x}_t] \rangle + \frac{L_1 \eta^2}{2} \mathbb{E}[\|\tilde{\mathbf{g}}_t\|^2 \mid \mathbf{x}_t].$$

Using (60), Young's inequality gives

$$-\langle \nabla f(\mathbf{x}_t), \mathbb{E}[\tilde{\mathbf{g}}_t \mid \mathbf{x}_t] \rangle \leq -\frac{3}{4} \|\nabla f(\mathbf{x}_t)\|^2 + \tilde{C}_b^{(R)} r^{4R+4}.$$

Combining this inequality with (61), and choosing $\eta \leq c/(L_1(V+1))$ for a sufficiently small constant $c > 0$, yields

$$\mathbb{E}[f(\mathbf{x}_{t+1}) \mid \mathbf{x}_t] \leq f(\mathbf{x}_t) - \frac{\eta}{2} \|\nabla f(\mathbf{x}_t)\|^2 + L_1 \eta^2 \tilde{C}_1 n \sigma_g^2 + \eta \tilde{C}_R r^{4R+4}.$$

Taking total expectation, summing over $t = 0, \dots, T-1$, and using $f(\mathbf{x}_T) \geq f^*$, we obtain

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\mathbf{x}_t)\|^2 \leq \frac{2(f(\mathbf{x}_0) - f^*)}{\eta T} + 2L_1 \eta \tilde{C}_1 n \sigma_g^2 + 2\tilde{C}_R r^{4R+4}.$$

Absorbing numerical constants into $\tilde{C}_1$ and $\tilde{C}_R$ gives (59). Since the minimum is bounded by the average, the proposition follows. $\square$

F. Experimental Details

F.1. Baseline Details

This appendix summarizes the baselines used in Section 5.1.

Two-point central difference. Given a direction $\mathbf{z} \sim \text{Unif}(\mathbb{S}^{n-1})$ or $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$, the two-point estimator is

$$\mathbf{g}_{2p}(\mathbf{x}; r, \mathbf{z}) = \frac{n}{2r} \left(f(\mathbf{x} + r\mathbf{z}) - f(\mathbf{x} - r\mathbf{z}) \right) \mathbf{z},$$

which uses $Q = 2$ function evaluations.

Kernel-based ZOGE. Sample $\mathbf{z} \sim \text{Unif}(\mathbb{S}^{n-1})$ and $\rho \sim \text{Unif}([-1, 1])$ independently, and compute

$$\mathbf{g}_{\text{ker}}(\mathbf{x}; r, \mathbf{z}, \rho) = \frac{n}{2r\rho} \left(f(\mathbf{x} + r\rho\mathbf{z}) - f(\mathbf{x} - r\rho\mathbf{z}) \right) k(\rho) \mathbf{z},$$

where we use the kernel

$$k(\rho) = \frac{195\rho}{64} \left(99\rho^4 - 126\rho^2 + 35 \right),$$

which yields $\mathcal{O}(r^6)$ bias (matching Romberg-ZOGE with $R = 2$). Each estimate uses $Q = 2$ function evaluations.

Unbiased ZOGE (P4). P4 is a randomized telescoping estimator. Given a probability mass function (PMF) $\{p_n\}_{n \geq 1}$ on $\mathbb{N}$, P4 draws a random level $N \sim \{p_n\}$ and returns

$$\mathbf{g}_{\text{P4}}(\mathbf{x}; r, \mathbf{z}) = \left(\frac{f(\mathbf{x} + r\mathbf{z}) - f(\mathbf{x})}{r} + \frac{1}{p_N} \left[\frac{f(\mathbf{x} + r_{N+1}\mathbf{z}) - f(\mathbf{x})}{r_{N+1}} - \frac{f(\mathbf{x} + r_N\mathbf{z}) - f(\mathbf{x})}{r_N} \right] \right) \mathbf{z},$$

which requires $Q = 4$ function evaluations per estimate.

Following (Ma & Huang, 2025b), we choose p_n to follow a Zipf distribution with exponent $s > 1$, i.e., $p_n = \frac{n^{-s}}{\zeta(s)}$ for $n = 1, 2, \dots$, where $\zeta(s) \triangleq \sum_{j=1}^{\infty} j^{-s}$. Given a base radius $r > 0$, we define a monotonically decreasing radius sequence $r_n n \geq 1$ using tail probabilities as $r_n = r \sum_{j=n}^{\infty} p_j = \frac{r}{\zeta(s)} \sum_{j=n}^{\infty} j^{-s}$, which is equivalently characterized by the recursion $r_n - r_{n+1} = r p_n$. In all experiments, we set $s = 1.5$, consistent with the setting used in (Ma & Huang, 2025b).

F.2. Synthetic details

We use $B = 120$ function evaluations per iteration throughout. Romberg-ZOGE with $R = 2$ uses $m = 20$ ($Q = 6$), Two-point and Kernel use $m = 60$ ($Q = 2$), and P4 uses $m = 30$ ($Q = 4$). For Poly8 we set $r = 10^{-3}$, and for Rosenbrock we set $r = 5 \times 10^{-4}$. All ZO-SGD results use 10 random initializations.

F.2.1. ADDITIONAL SYNTHETIC RESULTS UNDER GAUSSIAN DIRECTIONS

This section reports the counterparts of Section 5.2 under Gaussian directions $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$. We use the same protocol and hyperparameters as in Section 5.2, including $n = 100$, $M = 10$ evaluation points for bias/variance, $M_z = 10^7$ directions for approximating directional expectations, and 10 random initializations for ZO-SGD.

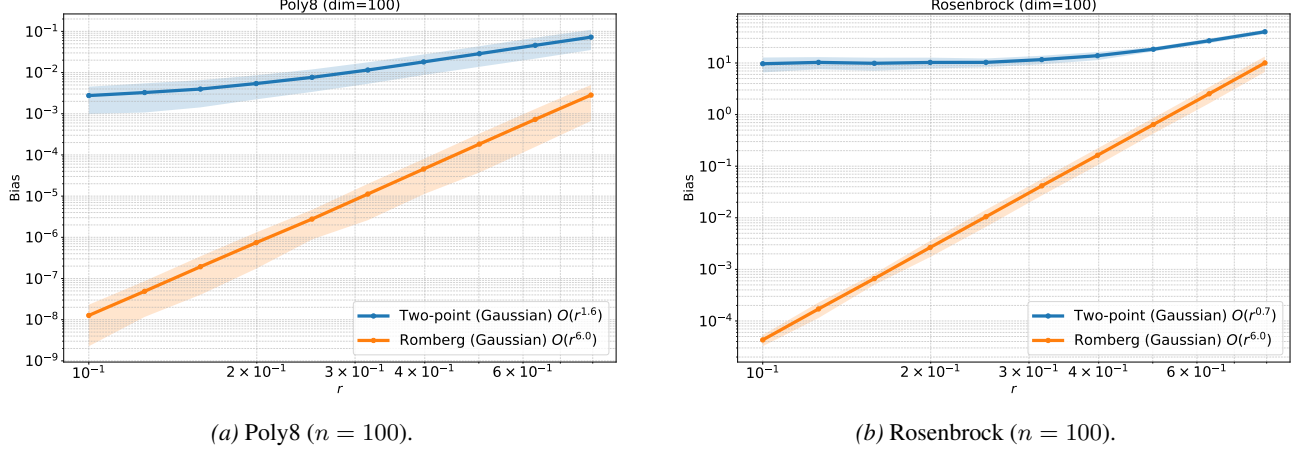


Figure 6. Bias versus smoothing radius r under $\mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$.

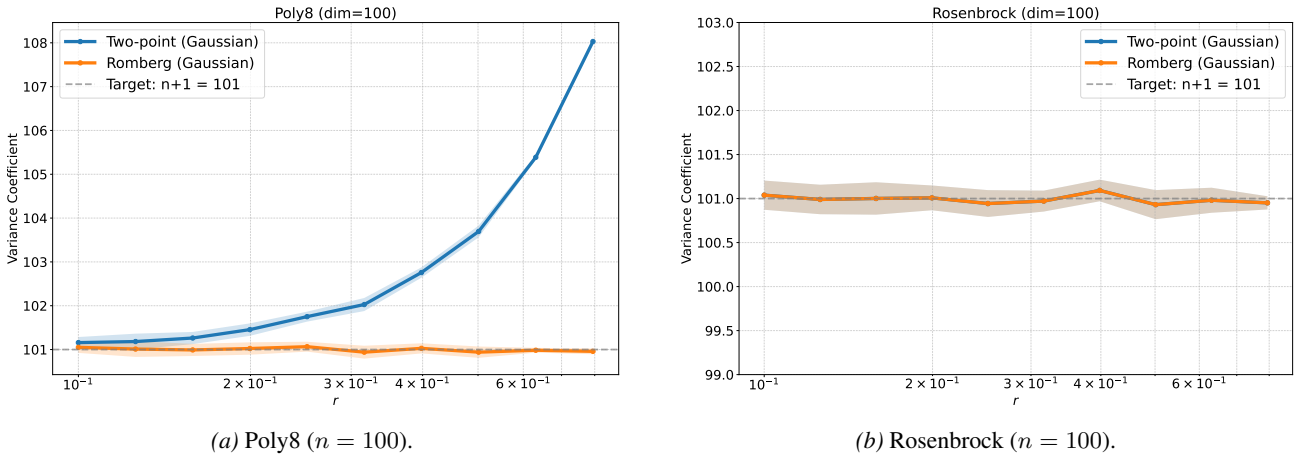


Figure 7. Directional variance coefficient $V(r)$ under $\mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$.

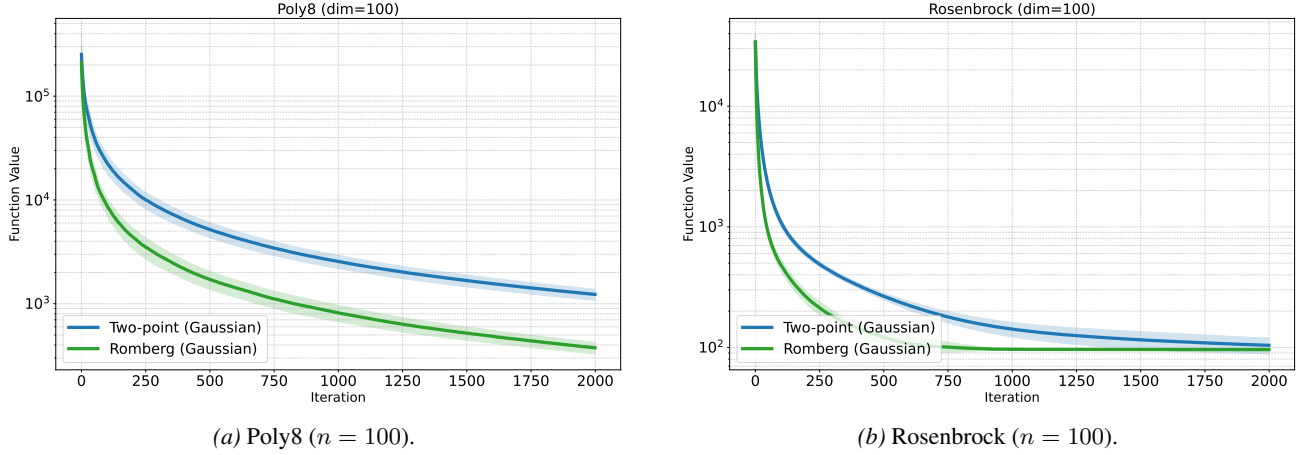


Figure 8. ZO-SGD under $\mathcal{N}(\mathbf{0}, \mathbf{I}_n/n)$ with the same number of function evaluations per iteration.

F.2.2. REFINED BIAS VALIDATION WITH AUXILIARY AVERAGING

In Figure 1, Kernel and P4 exhibit fitted slopes that do not perfectly match their nominal asymptotic behavior. This is mainly because these estimators involve auxiliary randomness in addition to the sampled direction $\mathbf{z}$, which can increase the Monte Carlo variance of the empirical bias estimate. To better isolate the asymptotic bias behavior, we conduct an additional refined bias-validation experiment.

We use the 10-dimensional Poly8 objective and perform the computation with `mpmath` using 50-digit precision. For Kernel and P4, which involve auxiliary randomness, we estimate the bias by averaging over both random directions and auxiliary random variables:

$$\text{Bias}_{\text{aux}}(r) \triangleq \frac{1}{M} \sum_{i=1}^M \left\| \frac{1}{M_z M_{\text{aux}}} \sum_{j=1}^{M_z} \sum_{\ell=1}^{M_{\text{aux}}} g(\mathbf{x}_i; r, \mathbf{z}_j, \xi_\ell) - \nabla f(\mathbf{x}_i) \right\|, \quad (62)$$

where ξ_ℓ denotes the auxiliary randomness of the estimator, such as ρ for Kernel and N for P4. In this experiment, we set $M_z = M_{\text{aux}} = 10^7$.

Table 2. Refined bias-validation results on the 10-dimensional Poly8 objective. The slope is obtained by a log-log fit over the displayed range of smoothing radii.

r	Romberg	Two-point	Kernel	P4
slope	6.00	2.01	6.07	1.21
1.00×10^{-1}	2.79×10^{-9}	3.86×10^{-4}	2.54×10^{-4}	4.92×10^{-11}
5.53×10^{-2}	7.98×10^{-11}	1.14×10^{-4}	7.86×10^{-6}	9.12×10^{-11}
3.06×10^{-2}	2.29×10^{-12}	3.45×10^{-5}	1.87×10^{-7}	3.05×10^{-11}
1.69×10^{-2}	6.55×10^{-14}	1.05×10^{-5}	4.92×10^{-9}	1.66×10^{-11}
9.36×10^{-3}	1.88×10^{-15}	3.22×10^{-6}	1.49×10^{-10}	1.50×10^{-11}
5.18×10^{-3}	5.38×10^{-17}	9.84×10^{-7}	4.23×10^{-12}	1.00×10^{-12}
2.87×10^{-3}	1.54×10^{-18}	3.01×10^{-7}	1.07×10^{-13}	1.09×10^{-12}
1.58×10^{-3}	4.42×10^{-20}	9.21×10^{-8}	3.16×10^{-15}	9.94×10^{-13}

As shown in Table 2, Two-point and Romberg-ZOG recover the expected $\mathcal{O}(r^2)$ and $\mathcal{O}(r^6)$ bias decay, respectively. Kernel also exhibits approximately $\mathcal{O}(r^6)$ decay under the refined auxiliary-averaging protocol. For P4, the empirical bias remains at a very low level, around 10^{-11} – 10^{-12} , rather than displaying a meaningful asymptotic slope. This is expected because P4 is theoretically unbiased, so the measured bias is dominated by the finite-sample fluctuation of the sample average, whose typical scale is $\sqrt{\frac{\text{Var}(g)}{M_z M_{\text{aux}}}}$, up to numerical precision. Since P4 also has a relatively large variance coefficient, this floor effect is more pronounced.

These refined results support the interpretation of Figure 1: the slope mismatch observed there should not be viewed as evidence against the theoretical bias properties of Kernel or P4. Rather, it is mainly due to finite-sample variance, auxiliary randomness, and numerical floor effects under the original Monte Carlo protocol.

F.2.3. SELECTED STEP SIZES.

We select the step size for each estimator using a short-horizon tuning procedure. For each candidate step size

$$\eta \in \{10^{-4}, 5 \times 10^{-5}, 10^{-5}, 5 \times 10^{-6}, 10^{-6}, 10^{-7}, 10^{-8}, 10^{-9}\},$$

We run ZO-SGD for $T_{\text{tune}} = 200$ iterations from 10 random initializations while keeping the number of function evaluations per iteration fixed, and compute the tail-averaged objective, defined as the average objective value over the final 50 iterations. We record the mean and standard deviation across successful runs, together with the number of successful runs succ . The step size η^* is chosen as the minimizer of the mean tail average among candidates with at least 80% successful runs (i.e., $\text{succ} \geq 8/10$). All reported optimization curves use the resulting η^* and are evaluated on the same set of initializations for all methods.

Based on Table 3, we use the following η^* in the main ZO-SGD experiments:

- **Poly8:** Romberg (S): $\eta^* = 5 \times 10^{-5}$; Romberg (G): $\eta^* = 5 \times 10^{-5}$; Two-point (S): $\eta^* = 10^{-5}$; Two-point (G): $\eta^* = 10^{-5}$; P4: $\eta^* = 10^{-7}$; Kernel: $\eta^* = 10^{-9}$.
- **Rosenbrock:** Romberg (S): $\eta^* = 5 \times 10^{-5}$; Romberg (G): $\eta^* = 5 \times 10^{-5}$; Two-point (S): $\eta^* = 10^{-5}$; Two-point (G): $\eta^* = 10^{-5}$; P4: $\eta^* = 10^{-5}$; Kernel: $\eta^* = 10^{-9}$.

F.2.4. EFFECT OF PARAMETERS ON ROMBERG-ZOGE

We examine the sensitivity of Romberg-ZOGE to key hyperparameters in ZO-SGD, including the step size η , the smoothing radius r , and the extrapolation order R . Unless otherwise stated, we fix two hyperparameters at their default values and vary the third. All curves are obtained while holding the number of function evaluations per iteration fixed, and are averaged over the same 10 random initializations as in the main experiments.

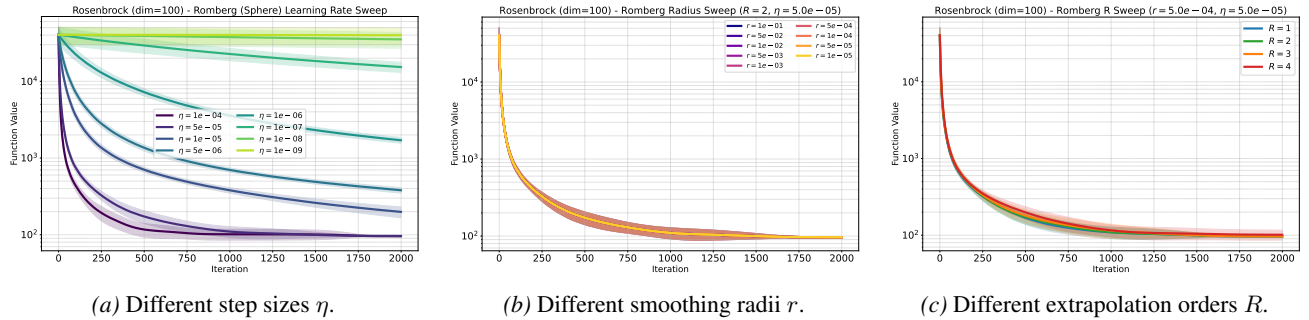


Figure 9. Sensitivity of Romberg-ZOGE in ZO-SGD on Rosenbrock ($n = 100$) under spherical directions. From left to right: varying the step size η , the smoothing radius r , and the extrapolation order R .

Discussion. We observe that the step size η primarily controls the rate of descent, whereas varying the smoothing radius r or the extrapolation order R does not qualitatively change the optimization trajectories. Romberg-ZOGE consistently makes stable progress across a broad range of parameter choices, suggesting strong robustness to hyperparameter tuning.

F.3. Noisy R -Sensitivity under Different Stochastic Formulations

We further study how the extrapolation order R affects optimization performance under stochastic oracles. The theory in Theorem 4.3 suggests a tradeoff: increasing R reduces the deterministic truncation bias, but under an independent noisy function-value oracle it also amplifies stochastic noise through the factor κ_R/r^2 . Thus, the preferred value of R is expected to depend on the oracle noise level.

Romberg-Extrapolated Zeroth-Order Gradient Estimator

Table 3. Step-size tuning via short-horizon tail-averaged objective ($T_{\text{tune}} = 200$, tail length = 50). Lower is better. Each entry reports **mean** $\pm$ **std** (top) and **succ**/10 (bottom), computed over successful runs.

η	Tail-averaged objective (Poly8)					
	Romberg (S)	Romberg (G)	Two-point (S)	Two-point (G)	P4	Kernel
5×10^{-4}	diverged (0/10)	diverged (0/10)	diverged (0/10)	diverged (0/10)	diverged (0/10)	diverged (0/10)
10^{-4}	diverged (0/10)	diverged (0/10)	$1.1\text{e}+2 \pm 0.0\text{e}+0$ (1/10)	$9.3\text{e}+1 \pm 0.0\text{e}+0$ (1/10)	diverged (0/10)	diverged (0/10)
5×10^{-5}	$2.6\text{e}+2 \pm 3.3\text{e}+1$ (8/10)	$2.6\text{e}+2 \pm 1.2\text{e}+1$ (8/10)	$2.7\text{e}+2 \pm 3.2\text{e}+1$ (7/10)	$2.6\text{e}+2 \pm 2.9\text{e}+1$ (7/10)	$2.6\text{e}+2 \pm 2.4\text{e}+1$ (3/10)	diverged (0/10)
10^{-5}	$1.5\text{e}+3 \pm 2.9\text{e}+2$ (10/10)	$1.4\text{e}+3 \pm 1.5\text{e}+2$ (10/10)	$1.3\text{e}+3 \pm 9.5\text{e}+1$ (10/10)	$1.3\text{e}+3 \pm 1.1\text{e}+2$ (10/10)	$1.4\text{e}+3 \pm 1.8\text{e}+2$ (5/10)	diverged (0/10)
5×10^{-6}	$3.0\text{e}+3 \pm 4.0\text{e}+2$ (10/10)	$3.0\text{e}+3 \pm 3.0\text{e}+2$ (10/10)	$2.9\text{e}+3 \pm 3.2\text{e}+2$ (10/10)	$2.9\text{e}+3 \pm 3.0\text{e}+2$ (10/10)	$3.0\text{e}+3 \pm 2.8\text{e}+2$ (7/10)	diverged (0/10)
10^{-6}	$1.1\text{e}+4 \pm 1.8\text{e}+3$ (10/10)	$1.1\text{e}+4 \pm 1.8\text{e}+3$ (10/10)	$1.1\text{e}+4 \pm 1.7\text{e}+3$ (10/10)	$1.1\text{e}+4 \pm 1.8\text{e}+3$ (10/10)	$1.1\text{e}+4 \pm 1.7\text{e}+3$ (7/10)	$3.4\text{e}+4 \pm 1.8\text{e}+3$ (3/10)
10^{-7}	$7.3\text{e}+4 \pm 2.4\text{e}+4$ (10/10)	$7.4\text{e}+4 \pm 2.5\text{e}+4$ (10/10)	$7.3\text{e}+4 \pm 2.4\text{e}+4$ (10/10)	$7.4\text{e}+4 \pm 2.5\text{e}+4$ (10/10)	$7.4\text{e}+4 \pm 2.5\text{e}+4$ (10/10)	$2.8\text{e}+5 \pm 2.7\text{e}+5$ (3/10)
10^{-8}	$2.3\text{e}+5 \pm 2.4\text{e}+5$ (10/10)	$2.3\text{e}+5 \pm 2.4\text{e}+5$ (10/10)	$2.2\text{e}+5 \pm 2.4\text{e}+5$ (10/10)	$2.3\text{e}+5 \pm 2.4\text{e}+5$ (10/10)	$2.3\text{e}+5 \pm 2.4\text{e}+5$ (10/10)	$5.2\text{e}+5 \pm 8.7\text{e}+5$ (5/10)
10^{-9}	$4.4\text{e}+5 \pm 4.4\text{e}+5$ (10/10)	$4.4\text{e}+5 \pm 4.4\text{e}+5$ (10/10)	$4.4\text{e}+5 \pm 4.4\text{e}+5$ (10/10)	$4.4\text{e}+5 \pm 4.4\text{e}+5$ (10/10)	$4.4\text{e}+5 \pm 4.4\text{e}+5$ (10/10)	$4.8\text{e}+5 \pm 5.0\text{e}+5$ (10/10)

η	Tail-averaged objective (Rosenbrock)					
	Romberg (S)	Romberg (G)	Two-point (S)	Two-point (G)	P4	Kernel
5×10^{-4}	diverged (0/10)	diverged (0/10)	diverged (0/10)	diverged (0/10)	diverged (0/10)	diverged (0/10)
10^{-4}	diverged (0/10)	diverged (0/10)	$3.9\text{e}+2 \pm 7.9\text{e}+1$ (7/10)	$3.7\text{e}+2 \pm 9.1\text{e}+1$ (6/10)	diverged (0/10)	diverged (0/10)
5×10^{-5}	$6.6\text{e}+2 \pm 1.1\text{e}+2$ (8/10)	$6.7\text{e}+2 \pm 1.1\text{e}+2$ (9/10)	$6.6\text{e}+2 \pm 1.1\text{e}+2$ (7/10)	$6.7\text{e}+2 \pm 9.1\text{e}+1$ (6/10)	$6.7\text{e}+2 \pm 7.9\text{e}+1$ (6/10)	diverged (0/10)
10^{-5}	$3.4\text{e}+3 \pm 2.3\text{e}+2$ (10/10)	$3.3\text{e}+3 \pm 2.2\text{e}+2$ (10/10)	$3.2\text{e}+3 \pm 1.7\text{e}+2$ (10/10)	$3.3\text{e}+3 \pm 1.7\text{e}+2$ (10/10)	$3.4\text{e}+3 \pm 1.6\text{e}+2$ (10/10)	diverged (0/10)
5×10^{-6}	$8.6\text{e}+3 \pm 3.7\text{e}+2$ (10/10)	$8.6\text{e}+3 \pm 4.1\text{e}+2$ (10/10)	$8.7\text{e}+3 \pm 4.3\text{e}+2$ (10/10)	$8.6\text{e}+3 \pm 4.5\text{e}+2$ (10/10)	$8.6\text{e}+3 \pm 4.6\text{e}+2$ (10/10)	diverged (0/10)
10^{-6}	$5.9\text{e}+4 \pm 7.4\text{e}+3$ (10/10)	$5.9\text{e}+4 \pm 7.3\text{e}+3$ (10/10)	$5.8\text{e}+4 \pm 6.5\text{e}+3$ (10/10)	$5.9\text{e}+4 \pm 7.0\text{e}+3$ (10/10)	$5.9\text{e}+4 \pm 7.7\text{e}+3$ (10/10)	$3.7\text{e}+5 \pm 0.0\text{e}+0$ (1/10)
10^{-7}	$3.4\text{e}+5 \pm 8.1\text{e}+4$ (10/10)	$3.4\text{e}+5 \pm 8.2\text{e}+4$ (10/10)	$3.4\text{e}+5 \pm 8.0\text{e}+4$ (10/10)	$3.4\text{e}+5 \pm 8.0\text{e}+4$ (10/10)	$3.4\text{e}+5 \pm 8.1\text{e}+4$ (10/10)	$5.1\text{e}+5 \pm 2.5\text{e}+5$ (3/10)
10^{-8}	$4.8\text{e}+5 \pm 2.2\text{e}+5$ (10/10)	$4.8\text{e}+5 \pm 2.2\text{e}+5$ (10/10)	$4.8\text{e}+5 \pm 2.2\text{e}+5$ (10/10)	$4.8\text{e}+5 \pm 2.2\text{e}+5$ (10/10)	$4.8\text{e}+5 \pm 2.2\text{e}+5$ (10/10)	$5.7\text{e}+5 \pm 2.5\text{e}+5$ (6/10)
10^{-9}	$5.8\text{e}+5 \pm 2.4\text{e}+5$ (10/10)	$5.8\text{e}+5 \pm 2.4\text{e}+5$ (10/10)	$5.8\text{e}+5 \pm 2.4\text{e}+5$ (10/10)	$5.8\text{e}+5 \pm 2.4\text{e}+5$ (10/10)	$5.8\text{e}+5 \pm 2.4\text{e}+5$ (10/10)	$5.8\text{e}+5 \pm 2.4\text{e}+5$ (10/10)

We evaluate this effect on the 100-dimensional Rosenbrock function. We use 10 random initializations, set $r = 5 \times 10^{-4}$, $\eta = 5 \times 10^{-5}$, and run ZO-SGD for $T = 200$ iterations. We report the final objective value as mean $\pm$ standard deviation.

We compare two stochastic formulations. In the independent noisy function-value oracle, each queried value uses an independent noise sample, i.e.,

$$\tilde{f}(\mathbf{x} \pm r_k \mathbf{z}; \boldsymbol{\xi}_{k,\pm}), \quad k = 0, \dots, R.$$

This is the setting analyzed in Theorem 4.3, where the κ_R/r^2 noise-amplification term appears. In the shared-sample stochastic-objective setting, the same sample is reused across all queried points within one Romberg estimator:

$$\tilde{f}(\mathbf{x} \pm r_k \mathbf{z}; \boldsymbol{\xi}), \quad k = 0, \dots, R.$$

As discussed in Remark 4.5, this shared-sample formulation removes the κ_R/r^2 amplification.

Table 4 shows the predicted order-selection transition under the independent noisy oracle. When the noise level is small, $R = 3$ performs best, reflecting the benefit of higher-order bias reduction. As σ increases, the preferred order shifts to $R = 2$

Table 4. Noisy R -sensitivity on the 100D Rosenbrock function under the independent noisy function-value oracle. Each entry reports the final objective value as mean $\pm$ standard deviation over 10 random initializations. Lower is better.

σ	$R = 0$	$R = 1$	$R = 2$	$R = 3$
10^{-3}	482.56 $\pm$ 72.42	482.43 $\pm$ 41.93	480.18 $\pm$ 57.33	417.83 $\pm$ 78.11
10^{-2}	496.60 $\pm$ 88.57	467.79 $\pm$ 75.37	465.01 $\pm$ 65.69	599.80 $\pm$ 79.53
10^{-1}	495.69 $\pm$ 78.53	532.32 $\pm$ 89.05	2858.89 $\pm$ 318.91	16590.11 $\pm$ 2785.00

Table 5. Noisy R -sensitivity on the 100D Rosenbrock function under the shared-sample stochastic-objective setting. Each entry reports the final objective value as mean $\pm$ standard deviation over 10 random initializations. Lower is better.

σ	$R = 0$	$R = 1$	$R = 2$	$R = 3$
10^{-3}	484.48 $\pm$ 62.79	482.70 $\pm$ 40.24	473.18 $\pm$ 95.88	434.05 $\pm$ 64.90
10^{-2}	482.13 $\pm$ 51.77	462.29 $\pm$ 86.33	458.08 $\pm$ 64.48	427.43 $\pm$ 67.06
10^{-1}	484.62 $\pm$ 73.23	456.25 $\pm$ 71.43	428.05 $\pm$ 71.88	424.68 $\pm$ 69.00

at $\sigma = 10^{-2}$ and further to $R = 0$ at $\sigma = 10^{-1}$, consistent with the increasing impact of the κ_R/r^2 noise-amplification term.

By contrast, Table 5 shows that the downward shift in the preferred order disappears under the shared-sample stochastic-objective formulation. In this setting, $R = 3$ remains effective across all tested noise levels. This behavior is consistent with the theory: when the same sample ξ is reused across the queried points within one Romberg estimator, the independent function-value noise accumulation is replaced by the usual stochastic-gradient variance contribution, and the κ_R/r^2 amplification no longer appears.

F.4. Wireless Network Optimization details

In this experiment, we set the smoothing radius to $r = 5 \times 10^{-4}$. Each run is initialized by a small random perturbation around the simulator’s default configuration, i.e., $\mathbf{x}_0$ is sampled in a neighborhood of the simulator-provided baseline $\bar{\mathbf{x}}_0$.

The wireless network performance evaluator used in our experiments is proprietary and cannot be released due to confidentiality constraints.

Smooth quantile objective. Given location-wise SE values $\{\eta^l(\mathbf{x})\}_{l=1}^L$ returned by the simulator, we define the (empirical) τ -quantile estimator via the pinball loss

$$\rho_\tau(c) \triangleq c(\tau - \mathbb{1}\{c < 0\}), \quad (63)$$

whose minimizer recovers the τ -quantile.

Following (Zhang et al., 2025), we use the smooth approximation

$$S_{\tau,\gamma}(c) \triangleq \tau c + \gamma \log(1 + \exp(-c/\gamma)), \quad \gamma > 0, \quad (64)$$

and define the smooth quantile operator as

$$\tilde{Q}_\tau(\{\eta^l(\mathbf{x})\}_{l=1}^L) \triangleq \arg \min_{z \in \mathbb{R}} \sum_{l=1}^L S_{\tau,\gamma}(\eta^l(\mathbf{x}) - z). \quad (65)$$

As $\gamma \rightarrow 0^+$, $S_{\tau,\gamma}$ approaches ρ_τ and $\tilde{Q}_\tau$ reduces to the empirical τ -quantile. We use $\tau = 0.9$, $\gamma = 0.1$, and the number of grids $L = 2048$.

Step-size selection. To ensure a fair comparison, we tune the step size of each method using the same short-horizon protocol as in Section F.2.3, while holding the number of function evaluations per iteration fixed.

We evaluate the same candidate grid $\eta \in \{10^{-4}, 5 \times 10^{-5}, 10^{-5}, 5 \times 10^{-6}, 10^{-6}, 10^{-7}, 10^{-8}, 10^{-9}\}$, run ZO-SGD for $T_{\text{tune}} = 200$ iterations from 10 random initializations, and compute the tail-averaged objective over the last 50 iterations. For each candidate η , we report mean $\pm$ std across successful runs together with the success count (succ/10) in Table 6. We

Table 6. Wireless surrogate step-size tuning via short-horizon tail-averaged objective ($T_{\text{tune}} = 200$, tail length = 50). Entries report mean $\pm$ std across successful runs with success counts (succ/10). Higher is better.

Step size η	Romberg (S)	Romberg (G)	Two-point (S)	Two-point (G)	P4	Kernel
10^{-4}	$1.9\text{e}+3 \pm 7.2\text{e}+0$ (10/10)	$1.9\text{e}+3 \pm 8.5\text{e}+0$ (10/10)	$1.9\text{e}+3 \pm 9.4\text{e}+0$ (10/10)	$1.9\text{e}+3 \pm 7.5\text{e}+0$ (10/10)	$1.9\text{e}+3 \pm 7.6\text{e}+0$ (10/10)	diverged (0/10)
5×10^{-5}	$1.9\text{e}+3 \pm 1.3\text{e}+1$ (10/10)	$1.9\text{e}+3 \pm 1.1\text{e}+1$ (10/10)	$1.9\text{e}+3 \pm 1.2\text{e}+1$ (10/10)	$1.9\text{e}+3 \pm 7.9\text{e}+0$ (10/10)	$1.9\text{e}+3 \pm 9.9\text{e}+0$ (10/10)	diverged (0/10)
10^{-5}	$1.8\text{e}+3 \pm 5.9\text{e}+0$ (10/10)	$1.8\text{e}+3 \pm 6.8\text{e}+0$ (10/10)	$1.8\text{e}+3 \pm 6.0\text{e}+0$ (10/10)	$1.8\text{e}+3 \pm 6.6\text{e}+0$ (10/10)	$1.8\text{e}+3 \pm 6.8\text{e}+0$ (10/10)	diverged (0/10)
5×10^{-6}	$1.8\text{e}+3 \pm 3.4\text{e}+1$ (10/10)	$1.8\text{e}+3 \pm 3.5\text{e}+1$ (10/10)	$1.8\text{e}+3 \pm 3.4\text{e}+1$ (10/10)	$1.8\text{e}+3 \pm 3.0\text{e}+1$ (10/10)	$1.8\text{e}+3 \pm 3.2\text{e}+1$ (10/10)	diverged (0/10)
10^{-6}	$1.6\text{e}+3 \pm 1.2\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 1.3\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 1.2\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 1.3\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 1.3\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 8.3\text{e}+0$ (1/10)
10^{-7}	$1.6\text{e}+3 \pm 9.9\text{e}+0$ (10/10)	$1.6\text{e}+3 \pm 6.2\text{e}+0$ (10/10)	$1.6\text{e}+3 \pm 9.9\text{e}+0$ (10/10)	$1.6\text{e}+3 \pm 6.2\text{e}+0$ (10/10)	$1.6\text{e}+3 \pm 6.2\text{e}+0$ (10/10)	$1.6\text{e}+3 \pm 6.0\text{e}+0$ (3/10)
10^{-8}	$1.6\text{e}+3 \pm 1.2\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 1.2\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 1.2\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 1.2\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 1.2\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 1.2\text{e}+1$ (10/10)
10^{-9}	$1.6\text{e}+3 \pm 8.9\text{e}+0$ (10/10)	$1.6\text{e}+3 \pm 1.3\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 8.9\text{e}+0$ (10/10)	$1.6\text{e}+3 \pm 1.3\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 1.3\text{e}+1$ (10/10)	$1.6\text{e}+3 \pm 1.3\text{e}+1$ (10/10)

select η^* as the minimizer of the mean tail average among candidates with at least 80% successful runs (i.e., $\text{succ} \geq 8/10$). The selected step sizes are: Romberg (S): $\eta^* = 10^{-4}$, Romberg (G): $\eta^* = 10^{-4}$, Two-point (S): $\eta^* = 10^{-4}$, Two-point (G): $\eta^* = 10^{-4}$, P4: $\eta^* = 10^{-4}$, and Kernel: $\eta^* = 10^{-8}$.

Reproducibility note. Although the exact simulator cannot be shared, the optimization pipeline (including estimator implementations, the protocol that fixes the number of function evaluations per iteration, and stopping criteria) is fully specified and matches that used in our reproducible benchmarks. Our conclusions are further supported by experiments in fully reproducible settings, including synthetic objectives (Section 5) and large-scale LLM fine-tuning (Section 5.4), which show trends consistent with the wireless network optimization results.

F.5. LLM fine-tuning details

In the LLM fine-tuning experiment, we compare different ZOGes under a unified ZO-SGD framework for black-box fine-tuning. We fine-tune a pretrained OPT-1.3B model on the SST-2 sentiment classification task. We use ZO-SGD throughout with learning rate $\eta = 10^{-4}$ and smoothing radius $r = 10^{-3}$, which is the same as (Malladi et al., 2023; Ma & Huang, 2025b). The batch-size of SGD is fixed at 16.

To ensure a fair comparison, we fix the total number of function evaluations per iteration to be $B = 12$ for all methods. Then the mini-batch size m for each baseline method are $m = 6$ for Two-point and Kernel, $m = 3$ for P4 and $m = 2$ for Romberg-ZOGE ($R = 2$).

All other hyperparameters follow standard black-box LLM fine-tuning setups, and the source code and reproduction instructions are provided in the supplementary material. Our implementation is adapted from the public black-box fine-tuning benchmark of (Zhang et al., 2024), with modifications to incorporate Romberg-ZOGE and to keep the number of function evaluations per iteration fixed across methods.